LENNOX

ENGINEERING DATA

WATER HEATING

# HM30/AM30

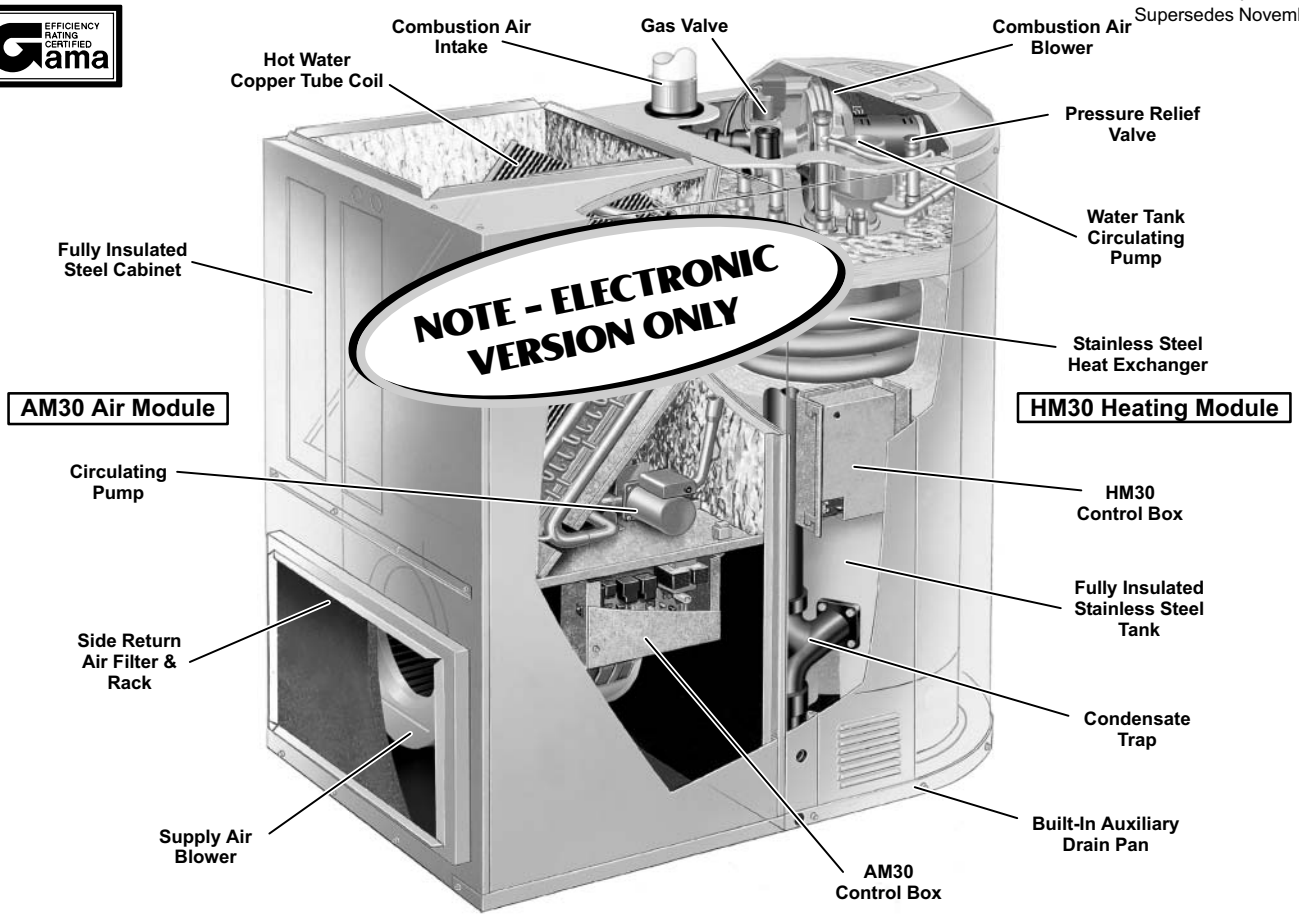
LENNOX CompleteHeat®

## SPACE HEATING/WATER HEATING SYSTEM

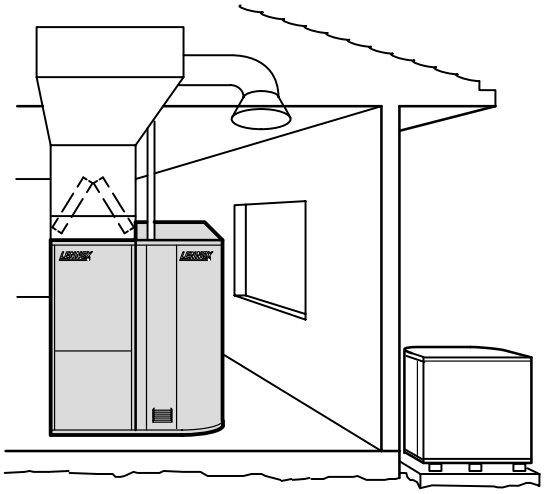
**CAE - 90.0%**

**Space Heating Input - 40,000 to 120,000 Btuh (11.7 to 35.2 kW)**
**Water Heating Input - 85,000 to 150, 000 Btuh (24.9 to 44.0 kW)**
**Nominal Add-On Cooling - 1-1/2 to 5 Tons (5.3 to 17.6 kW)**

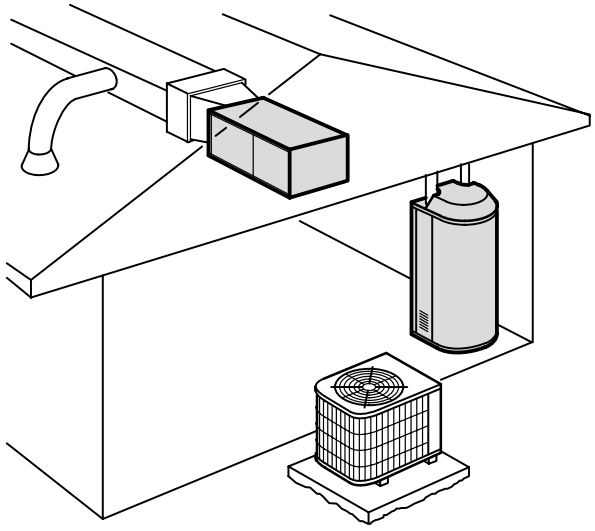
Bulletin No. 210057E
September 2002
Supersedes November 2001




### Typical Applications



Up-Flow Close-Coupled Installation with Cooling Coil



Horizontal Remotely Located Installation with Cooling Coil

NOTE - Due to Lennox' ongoing committment to quality, Specifications, Ratings and Dimensions subject to change without notice and without incurring liability.
Improper installation, adjustment, alteration, service or maintenance can cause property damage or personal injury.
Installation and service must be performed by a qualified installer and servicing agency.

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## FEATURES

**Application**
- Lennox CompleteHeat combination forced air and water heating system combines high efficiency space heating with high efficiency water heating.
- CompleteHeat system consists of HM30 Water Heating Module and AM30 Air Module.
- HM30 available for use with natural or LPG/Propane gas.
- HM30 tank stores 30 US gallons (114 L) of hot water at adjustable temperatures ranging from 110°F to 160°F (43° to 71.1°C).
- HM30 is capable of operating as a stand-alone water heater module as well as part of the CompleteHeat system. HM30 may also be used with radiant heating systems.
- AM30 may be used with other makes of water heaters.
- AM30 circulates hot water through a hot water coil, supply air blower extracts the heat from the coil and distributes heated air throughout the conditioned space.
- AM30 is a multi-position (up-flow, down-flow or horizontal) blower/hot water coil unit.
- Lennox add-on evaporator coil with remote condensing unit, electronic air cleaner and automatic humidifier can easily be added for a complete all season system.

**Equipment Warranty**
- HM30 stainless steel heat exchanger/hot water tank - limited ten year warranty in residential applications.
- All other covered components - limited five year warranty in residential applications.
- Refer to Lennox Equipment Limited Warranty certificate included with equipment for details.

**Approvals**
- Units certified by A.G.A./C.G.A. Laboratories and ratings are certified by GAMA.
- Approved by California Energy Commission and meet California Seasonal Efficiency requirements and California Nitrogen oxides ($NO_x$) Standards.
- Blower data from unit tests conducted in Lennox Laboratory air test chamber.
- HM30 units approved for vertical or horizontal (sidewall) venting.
- manufactured in accordance with ISO 9002 quality standards.

**Sequence of Operation**
- HM30 Heating Module uses potable tap water and heats it to a predetermined temperature with a gas power burner and helical heat exchanger.
- Thermistor controls burner operation, keeping water at preselected temperature.
- Stainless steel tank stores water until there is demand for domestic hot water or space heating.
- AM30 Air Module includes supply air blower, water circulating pump, hot water coil, and blower/pump control.
- When demand for space heat is received from the thermostat, the blower/pump control simultaneously activates the circulating pump and sends a heating demand signal to the HM30 control which automatically operates at high or low gas input depending on water tank temperature.
- After timed-on delay (adjustable) the blower energizes at heating speed.
- When heating demand is satisfied, the circulating pump shuts off.
- After a timed-off delay (fixed) the supply air blower shuts off.
- AM30 unit will shut off if HM30 tank temperature falls below 20° of HM30 tank temperature setting, giving priority to domestic water heating system.

## FEATURES - HM30 HEATING MODULE

**Cabinet**
- Heavy gauge steel construction.
- HM30 controls accessible from front of cabinet.
- Controls easily relocated from front of cabinet to the back for easy access. See dimension drawing.
- All plumbing connections located at top of cabinet.
- Auxiliary base drain pan integral part of cabinet. Meets local code requirements in certain areas.

**Tank Assembly**
- Stainless steel construction.
- Tank sides and top fully insulated (foam on sides, fiberglass on top).
- Water connections located on top of tank.
- Built-in pump circulates water to prevent stratification and sediment build-up.

**Tank Drain Valve**
- Furnished for servicing tank.
- Located behind access panel.
- Accessible from either side of cabinet.
- Standard garden hose connection.

**Lennox Designed Flue Pipe/Condensate Trap Assembly**
- Vents combustion products and collects condensate.
- Pipe connects to flue with one-piece no hub connector.
- Contains a built-in internal trap and removable boot on bottom for easy cleaning and servicing.

**Heat Exchanger**
- Stainless steel construction.
- Helical design.

**Burner Assembly**
- Stainless steel construction.
- Short flame design.
- Uses 100% outside air for combustion.

**Automatic Gas Control**
- Combines automatic electric valve (dual) and gas pressure regulation into a compact combination control.
- Dual valve design provides double assurance of close off of gas to main burners at each off cycle.
- Gas valve is automatically regulated to maintain even gas flow regardless of venting installation type.
- Design compensates for variations in gas supply pressure.

**FEATURES - Continued on Next Page ▶**

## FEATURES - HM30 HEATING MODULE - CONTINUED

**Integrated Water Temperature/Direct Spark Ignition Module**
- Module factory installed in HM30 control box.

**Ignition Control**
- Provides positive ignition of burner on each operating cycle.
- Main burner is extinguished during off cycle.
- Burner operation is automatic on demand for heat.
- Unit shuts down if loss of flame occurs and will initiate 4 attempts at relighting before going into Watchguard mode.
- Watchguard circuit automatically resets ignition controls after one hour of unit lockout. Eliminates nuisance calls for service.
- Diagnostic LED's are furnished as aid in troubleshooting.

**Water Temperature Control**
- Integral part of control module.
- Easy to read dial type.
- Adjustable 110°F to 160°F (43° to 71.1°C).

**Dual Limit Controls**
- Primary (manual reset) switch and secondary (auto-reset) switch.
- Both located on top of water tank.

**Combustion Air Blower**
- Two stage operation dependent on tank temperature.
- Aluminum housing, completely sealed.
- Pressure switch prevents unit operation in case of combustion air, flue outlet or condensate drain blockage.
- Located on top of water tank for easy access.

**Water Pressure Reducing Valve**
- 3/4 in. valve reduces inlet water pressure.
- Factory set at 55 psig (379 kPa).
- Supplied with unit for field installation.

**Expansion Tank**
- 3/4 in. male NPT, installed on the supply side of the HM30.
- Prevents excessive build up of system pressure due to thermal expansion.
- Protects system components from premature failure caused by pressure cycling.
- Supplied with unit for field installation.

**Temperature/Pressure Relief Valve**
- Provides temperature/pressure relief in case of abnormal operating conditions.
- Located on top of water tank.
- Opens at 210°F or 150 psig (99°C or 1034 kPa).
- Approved by A.S.M.E.

## HM30 OPTIONAL ACCESSORIES

**Thermostat**
- See Thermostats bulletin in Thermostats and Controls section and Lennox Price Book.

**Stand-Alone Transformer Kit (78J43)**
- 120/24 volt 40VA transformer and junction box allows HM30 to operate as a stand-alone high efficiency water heater.

**Water Mixing Valve Kit**
- Field adjustable to ensure a steady, safe temperature for domestic hot water.
- Recommended in applications using over 120°F (32°C) water tank temperature.
- Required in applications using over 140°F (60°C) water tank temperature or maximum temperature allowed by local codes.
- See Specifications table for kits available and ordering information.

**HM30 Control Interface Kit (90J03)**
- Allows HM30 to be used with radiant heating and other hot water coils.
- Contains control board in junction box to operate pump and system (similar to controls found in AM30).

**HM30 Heating Zone Control Board (90J02)**
- Solid-state control board with junction box operates up to four AM30 units in heating mode with one HM30 unit for zoning applications.

**Water Pressure Test Gauge (56L50)**
- Testing of water pressure supplied within a building.
- 3/4 in. bib hose connection.
- Second gauge needle indicates maximum water pressure reached.

**Air Vent/Bleed Valve**
- Available in 3/4 in. or 1 in. female NPT.
- Reduces air entrapped in system.
- See Specifications table for ordering information.

**Intake/Exhaust Vent Adaptor Kit**
- 3 in. to 2 in. (76 mm to 51 mm) reducer adaptors allows connection of exhaust and intake pipe to HM30.
- See Specifications table for ordering information.

**OPTIONAL ACCESSORIES - Continued on Next Page ▶**

## HM30 OPTIONAL ACCESSORIES

**Intake Air Filter Kit (28L82)**
- Helps keep debris out of the intake air stream.
- Recommended installation above the no hub connector in the vertical position.
- Includes one each 2 in. and 3 in. (51 mm and 76 mm) reducers.

**Concentric Termination Kit**
- Facilitates installation of combustion air intake pipe and flue exhaust pipe.
- 1-1/2 inch (38 mm) kit LB-49107CE **(60G77)** or 2 inch (51 mm) kit (**33K97**) contains concentric termination assembly, mounting clamp, roof flashing, reducer bushing and 45 degree elbow.
- Kit requires single hole penetration of roof or wall for installation.
- AGA/CGA certified.
- See Specifications table and dimension drawings.

**Roof Termination Kit**
- Facilitates installation of combustion air intake pipe and flue exhaust pipe.
- 2 inch (51 mm) kit, LB-49107CC **(15F75)**, or 3 inch (76 mm) kit, LB-65678A **(44J41)**, contains two neoprene rubber roof flashings.
- See Specifications table and dimension drawings.
- Refer to venting tables in this bulletin to determine pipe size needed and proper termination kit required.

**Wall Assembly Termination Kits**
- Facilitates installation of combustion air intake pipe and flue exhaust pipe.
- Refer to venting tables in this bulletin to determine pipe size needed and proper termination kit required.
- See Specifications table and dimension drawings.

**Close Couple Kits**
- **(22G44)** 2 inch (51 mm) or **(44J40)** 3 inch (76 mm) consists of close-couple side-by-side PVC piping with galvanized steel wall cover plate for sealing and isolating piping penetration of the wall.
- Piping spacing and length is sized for proper wall installations.
- AGA/CGA certified.

**WTK Close Couple Kits**
- 2 in. (51 mm) **(30G28)** or 3 in. (76 mm) **(81J20)** contains one insulated faceplate, one insulated exhaust pipe, elbow and fittings.

**Wall Ring Kit (15F74)**
- 2 inch (51 mm) contains 2 stainless steel outside seal caps, 2 galvanized steel inside seal caps, 4 seal rings for the caps and 18 inch (457 mm) insulation sleeve for sealing and isolating intake and exhaust piping penetration of wall.
- Maintain a maximum of 6 inches (152 mm) between the inlet and outlet openings in the installation of the pipes.

**WTKX Close Couple Kit With Extension Riser (30G79)**
- 2 inch (51 mm) is used where extended grade line clearance is required, not for 3 in. (76 mm) venting.
- Includes 3 ft. (1.0 m) extension riser containing both vent lines (exhaust vent insulated) and wall securing bracket.

**Condensate Drain Heat Cable Kits**
- Self-limiting wattage heat cable prevents condensate drain from freezing in unconditioned areas.
- Heat cable kits are available in the following lengths: 50 ft. (15.2 m) - LB-88643C **(26K70)**, 24 ft. (7.3 m) - LB-88643B **(26K69)** or 6 ft. (1.8 m) - LB-88643A **(26K68)**
- Also available Heat Cable Tape: 66 ft. (20 m) length, 1/2 in. (13 mm) wide fiberglass **(39G04)** or 60 ft. (18 m) length, 2 in. (51 mm) wide aluminum foil **(39G03)**.

## FEATURES - AM30 AIR MODULE

**Cabinet**
- Heavy gauge steel construction with five station metal wash process and baked-on enamel paint finish.
- Foil faced fiberglass insulation on hot water coil access door panel, side panels and on back panel reduces cabinet temperatures.
- Black mat faced fiberglass in blower compartment assures quiet operation.
- Complete service access by removing hot water coil section and blower compartment access panels.
- Blower assembly and filter completely removable for servicing.
- Blower deck rails angle down for easy blower removal.
- Electrical inlets in both sides of the cabinet.
- Return air entry on either side or bottom of cabinet. Optional on rear. Perforated knockout allows bottom return air.
- AM30 air module is applicable to up-flow, horizontal or down-flow installations.

**24 Volt Transformer**
- Furnished in AM30 control box.
- Operates both AM30 and HM30 controls.

**Field Make-up Box**
- Furnished for line voltage wiring.
- Box may be installed on either side of AM30 cabinet.
- Contains plug-in connection for power supply wiring, wire for 120 volt (less than 4 amps) accessory connection and all necessary hardware for installation.

**Water Heating Coil**
- Corrugated/lanced aluminum fins.
- Seamless corrosion resistant copper tubes.
- Fin collars grip tubing for maximum contact area.
- Flared shoulder tubing connections and silver soldering provides tight, leakproof joints.
- Factory tested under high pressure.
- Entire coil accessible for cleaning.

**Blower**
- Multi-speed direct drive blower.
- Statically and dynamically balanced.
- Blower motor resiliently mounted.
- Blower speed selection accomplished by simple wiring change. See Blower Performance tables.

**FEATURES - Continued on Next Page ▶**

## FEATURES - AM30 AIR MODULE (CONTINUED)

**Cleanable Air Filter**
- Washable or vacuum cleanable polyurethane frame type filter.

**Up-Flow, Horizontal or Down-Flow Bottom Return Air Applications**
- Filter secured in cabinet by fixed rear filter clip and two field installed side filter clips.

**Up-Flow Rear Return Air Applications**
- Filter must be field provided.
- See AM30 dimension drawing for field fabricated opening in rear of unit cabinet.

**Up-Flow Side Return Air Applications**
- External filter rack furnished with AM30 for side return air applications.
- Filter access door.
- Uses filter furnished with AM30.
- Flanges furnished for duct connection.
- See dimension drawing for details.

**AM30 Blower/Pump Control Board**
- Furnished in AM30 control box.
- Blower “timed-on” delay jumper (15 to 60 seconds adjustable) factory setting 15 seconds, blower “timed-off” delay setting 30 seconds (fixed).
- Built-in continuous low speed blower function.
- Automatically circulates water for 30 seconds every 6 hours (delays pump operation during cooling demand).
- Control voltage “plug-in” type terminal strip for thermostat connections.
- 120 volt (less than 4 amps) accessory terminals furnished for humidifiers and electronic air cleaners.
- Blower cooling relay for air-conditioning operation.
- Metal oxide varistor, mounted on board, protects against voltage spikes.
- Diagnostic LED furnished as an aid in troubleshooting.

**Freezestat**
- Protects system from freezing temperatures when unit is installed in unconditioned areas.
- Thermostat automatically energizes circulating pump when water line temperature falls below 45°F (7°C).

**Circulating Pump**
- Bronze construction.
- Carbon bearings.
- Self lubricating.
- Non-metallic impeller.
- Impedance protected.
- Motor and impeller removable as single unit for servicing.

## AM30 OPTIONAL ACCESSORIES

**Auxiliary Pump**
- For remote applications over 30 ft. (9.1 m).
- Field installed anywhere in-between AM30 and HM30 (see piping schematic).
- See Specifications table for ordering information.

**Anti-Thermal Siphon Kit (73J84)**
- In-line check valve prevents thermal siphoning when AM30 is located above HM30.

**Down-flow/Horizontal Kit (74J25)**
- Converts AM30 unit from up-flow to down-flow or horizontal applications. Kit includes all required water piping elbows for AM30 connections. Straight piping not included.

**Furnace Twinning Kit**
- Field installed kit (**15L38**) is required to operate two units simultaneously.
- Kit consists of twinning control and two fan sensors.
- Twinning kit requires Heating Zone Control Board **(90J02)** if HM30 serves two AM30’s.

**Horizontal Support Frame Kit (56J18)**
- Provides support of AM30 unit in horizontal applications.
- (2) 1 x 1-1/2 x 32-5/8 in. (25 x 38 x 829 mm) and (2) 1 x 3 x 53-7/8 in. (25 x 76 x 1368 mm) painted, heavy gauge cold rolled steel support channels with assembly and suspending holes furnished.
- Bolts and nuts for assembly furnished.

## INSTALLATION CLEARANCES

| | HM30 Heating Module | AM30 Up-flow / Down-flow | AM30 Horizontal |
|---|---|---|---|
| Vent Type | Plastic (PVC or ABS) | - - - - | |
| Sides | | 0 inches (0 mm) | |
| Rear | | 0 inches (0 mm) | |
| Top | 1 inch (25 mm) | 0 inches (0 mm) | **0 inches (0 mm) |
| Top and Front (service) | 30 inches (762 mm) | - - - - | |
| *Front | | *0 inches (0 mm) | |
| Floor | | Combustible | 0 inches (0 mm) |
| Exhaust Pipe | 0 inches (0 mm) | - - - - | |
| Exhaust Pipe (service) | 6 inches (152 mm) | - - - - | |
| Service Clearance (condensate drain) | 3 inches (76 mm) | - - - - | |

NOTE-Termination location must conform to the methods outlined in American National Standard (ANSI-Z223.1) National Fuel Gas Code or National Standard of Canada CAN/CGA-149.1, and CAN/CGA-149.2 “Installation Code for Gas Burning Appliances”.
*Front clearance for alcove installations is 24 inches (610 mm).
**Line contact installation permissible between jacket top or sides and building joists.

## SPECIFICATIONS - HM30 HEATING MODULE

| Heat Module Model No. | | HM30-100 | HM30-150 |
|---|---|---|---|
| Input - Btuh (kW) high | | 100,000 (29.3) | 150,000 (44.0) |
| Input - Btuh (kW) low | | 85,000 (24.9) | 130,000 (38.1) |
| Output - Btuh (kW) high - space heating | | 90,000 (26.4) | 135,000 (39.6) |
| Output - Btuh (kW) low - space heating | | 76,500 (22.4) | 117,000 (34.3) |
| Output - Btuh (kW) high - water heating | | 94,000 (27.5) | 141,000 (41.3) |
| Output - Btuh (kW) low - water heating | | 79,900 (23.4) | 122,200 (35.8) |
| First Hour Rating - U.S. gals (L) | | 163 (617) | 217 (821) |
| *$CA_{ef}$ (Effective Water Heating Efficiency) | | .86 | .86 |
| Recovery Efficiency | | 94% | 94% |
| Rated storage volume - U.S. Gallons (L) | | 32 (121) | 30 (114) |
| Recovery rate at 90°F (32°C) temperature rise- U.S. gals/hr. (L/hr.) | | 125 (473) | 188 (712) |
| Intake pipe size connection (PVC) - in. (mm) | | 2 (51) | 2 (51) |
| Exhaust pipe size connection (PVC) - in. (mm) | | 2 (51) | 2 (51) |
| Water connections - inlet/outlet I.D. - in. (mm) | Connections to water supply (fpt) | 3/4 (19.1) | 3/4 (19.1) |
| | Connections to AM30 Air Module (sweat) | 1 (25.4) | 1 (25.4) |
| Condensate trap drain connection (mpt) - in. (mm) | | 1/2 (12.7) | 1/2 (12.7) |
| Tank drain connection (standard garden hose connection) (12 tpi) - in. (mm) | | 1-1/16 (27) | 1-1/16 (27) |
| Gas Piping Size I.P.S in. (mm) | Natural Gas | 1/2 (12.7) | 3/4 (19.1) |
| | LPG/propane | 1/2 (12.7) | 1/2 (12.7) |
| Temperature/Pressure Relief Valve (furnished) | Opening specifications | 210°F or 150 psig (99°C or 1034 kPa) | |
| | Connection size - in. (mm) | 3/4 (19.1) | 3/4 (19.1) |
| Water Pressure Regulator | | 3/4 in. NPT - factory setting 55 psig (379 kPa) | |
| Expansion Tank | | 3/4 in. inlet | 3/4 in. inlet |
| [1] Shipping weight - lbs. (kg) 2 packages | | 191 (87) | 215 (98) |
| Electrical characteristics | | 120 volts - 60 hertz - 1 phase | |

### OPTIONAL ACCESSORIES (MUST BE ORDERED EXTRA)

| | | | |
|---|---|---|---|
| Concentric Termination Kits | 1-1/2 inch (38 mm) | **60G77** | **60G77** |
| | 2 inch (51 mm) | **33K97** | **33K97** |
| †Roof Termination Kits | 2 inch (51 mm) | **15F75** | **15F75** |
| | 3 inch (76 mm) | **44J41** | **44J41** |
| †Wall Assembly Termination Kits | 2 inch (51 mm) | **15F74** (wall ring kit), **22G44** (close couple) **30G28** (WTK close couple) **30G79** (WTKX close couple with extension riser) | |
| | 3 inch (76 mm) | **44J40** (close couple) - **81J20** (WTK close couple) | |
| Air Intake Filter Kit | | **28L82** | **28L82** |
| Intake/Exhaust Vent Adaptor Kit - 3 inch (76 mm) to 2 inch (51mm) | | **78J39** | **78J39** |
| Water Pressure Test Gauge | | **56L50** | **56L50** |
| Stand-alone Transformer Kit | | **78J43** | **78J43** |
| Water Mixing Valve Kits | 3/4 inch (19 mm), adjustable 85°F to 150°F (29°C to 66°C) | **99L99** | **99L99** |
| | 1 inch (25 mm), adjustable 85°F to 150°F (29°C to 66°C) | **10M00** | **10M00** |
| Air Vent/ Bleed Valve | 3/4 inch (19 mm) | **29K49** | **29K49** |
| | 1 inch (25.4 mm) | **29K50** | **29K50** |
| HM30 Heating Zone Control Board | | **90J02** | **90J02** |
| HM30 Control Interface Kit | | **90J03** | **90J03** |
| Condensate Drain Heat Cable | | **26K68** 6 ft. (1.8 m) - **26K69** 24 ft. (7.3 m) - **26K70** 50 ft. 15.2 m | |
| Heat Cable Tape - 1 roll | | **39G04** 1/2 in. (38 mm) fiberglass - **39G03** 2 in. (25 mm) aluminum foil | |

*$CA_{ef}$ (Effective Water Heating Capacity) = The effective efficiency of the combined appliance in performing the function of providing potable hot water.
**NOTE - $CA_{ef}$ is the same rating as the Energy Factor (EF) for water heaters as determined by U.S. Department of Energy test procedures.**
†Determine from venting tables proper intake and exhaust pipe size and termination kit required.
[1] Package one is unit, package two is expansion tank.

## SPECIFICATIONS - AM30 AIR MODULE

| Air Module Model No. | | | AM30Q2-40 | AM30Q2/3-70 | AM30Q3/4-70 |
|---|---|---|---|---|---|
| *Nominal heating capacity - Btuh (kW) | | | 40,000 (11.7) | 70,000 (20.5) | 70,000 (20.5) |
| **Temperature rise range - °F (°C) | | | 32 - 73 (18 - 41) | 35 - 78 (19 - 43) | 39 - 71 (22 - 39) |
| Blower wheel nominal diameter x width | | in. | 10 x 7 | 10 x 8 | 10 x 10 |
| | | mm | 254 x 178 | 254 x 203 | 254 x 254 |
| Blower motor output - hp (W) | | | 1/5 (149) | 1/3 (224) | 1/2 (373) |
| Circulating pump | Motor output - hp (W) | | 1/40 (19) | 1/40 (19) | 1/40 (19) |
| | †Capacity - U.S. Gals per minute (L per minute) | | 6 (23) | 6 (23) | 6 (23) |
| Heating Coil | Heating capacity range | Btuh | 21,600 - 55,300 | 27,500 - 90,000 | 43,800 - 109,400 |
| | | kW | 6.3 - 16.2 | 8.1 - 26.4 | 12.8 - 32.1 |
| | Net face area - sq. ft. ($m^2$) | | 3.83 (.36) | 3.83 (.36) | 4.33 (.40) |
| | Tube diameter - in. (mm) no. of rows | | 3/8 (9.5) - 1 | 3/8 (9.5) - 2 | 3/8 (9.5) - 2 |
| | Fins per inch (m) | | 16 (630) | 16 (630) | 16 (630) |
| | Water line connections (sweat) - in. (mm) I.D. | inlet | 3/4 (19) | 3/4 (19) | 1 (25.4) |
| | | outlet | 3/4 (19) | 3/4 (19) | 1 (25.4) |
| ★Number and size of filters | | in. | (1) 14 x 25 x 1 | (1) 14 x 25 x 1 | (1) 20 x 25 x 1 |
| | | mm | (1) 356 x 635 x 25 | (1) 356 x 635 x 25 | (1) 508 x 635 x 25 |
| Nominal cooling that can be added | | Tons | 1 thru 2 | 1 thru 3 | 2 thru 4 |
| | | kW | 3.5 thru 7.0 | 3.5 thru 10.6 | 7.0 thru 14.1 |
| Shipping weight - lbs. (kg) 1 package | | | 127 (58) | 144 (65) | 157 (71) |
| Electrical characteristics | | | 120 volts - 60 hertz - 1 phase | | |
| **OPTIONAL ACCESSORIES (MUST BE ORDERED EXTRA)** | | | | | |
| Auxiliary Pump | | | **99K69** | **99K69** | **99K69** |
| Anti-Thermal Siphon Kit | | | **73J84** | **73J84** | **73J84** |
| Downflow/Horizontal Kit | | | **74J25** | **74J25** | **74J25** |
| Horizontal Support Frame Kit | | | **56J18** | **56J18** | **56J18** |
| Twinning Kit (Requires Heating Zone Control Board, see HM30 Optional Accessories) | | | **15L38** | **15L38** | **15L38** |

*At 140°F (60°C) entering water temperature, 65°F (18°C) entering air temperature.
**120°F (49°C) High Speed or 170°F (77°C) Low Speed.
★Polyurethane frame type filter.
†140°F (60°C) water temperature.

## SPECIFICATIONS - AM30 AIR MODULE

| Air Module Model No. | | | AM30Q3/4-90 | AM30Q4/5-90 | AM30Q3/4-105 | AM30Q4/5-120 |
|---|---|---|---|---|---|---|
| *Nominal heating capacity - Btuh (kW) | | | †90,000 (26.4) | †90,000 (26.4) | †105,000 (30.8) | 120,000 (35.2) |
| Temperature rise range - °F (°C) | | | 35 - 76 (19 - 42) | 31 - 68 (17 - 38) | 41 - 88 (23 - 49) | 38 - 80 (21 - 44) |
| Blower wheel nominal diameter x width | | in. | 10 x 10 | 12 x 9 | 10 x 10 | 12 x 9 |
| | | mm | 254 x 254 | 305 x 229 | 254 x 254 | 305 x 229 |
| Blower motor output - hp (W) | | | 1/2 (373) | 3/4 (560) | 1/2 (373) | 3/4 (560) |
| Circulating pump | Motor output - hp (W) | | 1/25 (30) | 1/25 (30) | 1/25 (30) | 1/25 (30) |
| | †Capacity - U.S. Gals per minute (L per minute) | | 9.5 (36) | 9.5 (36) | 9.5 (36) | 9.5 (36) |
| Heating Coil | Heating capacity range | Btuh | 46,600 - 120,000 | 58,000 - 136,000 | 53,500 - 142,600 | 68,000 - 163,700 |
| | | kW | 13.7 - 35.2 | 17.0 - 39.8 | 15.7 - 41.8 | 19.9 - 48.0 |
| | Net face area - sq. ft. ($m^2$) | | 4.33 (.40) | 4.33 (.40) | 4.33 (.40) | 4.33 (.40) |
| | Tube diameter - in. (mm) no. of rows | | 3/8 (9.5) - 2 | 3/8 (9.5) - 2 | 3/8 (9.5) - 3 | 3/8 (9.5) - 3 |
| | Fins per inch (m) | | 16 (630) | 16 (630) | 16 (630) | 16 (630) |
| | Water line connections (sweat) - in. (mm) I.D. | inlet | 1 (25.4) | 1 (25.4) | 1 (25.4) | 1 (25.4) |
| | | outlet | 1 (25.4) | 1 (25.4) | 1 (25.4) | 1 (25.4) |
| ★Number and size of filters | | in. | (1) 20 x 25 x 1 | (1) 20 x 25 x 1 | (1) 20 x 25 x 1 | (1) 20 x 25 x 1 |
| | | mm | (1) 508 x 635 x 25 | (1) 508 x 635 x 25 | (1) 508 x 635 x 25 | (1) 508 x 635 x 25 |
| Nominal cooling that can be added | | Tons | 2 thru 4 | 3.5 thru 5 | 2 thru 4 | 3.5 thru 5 |
| | | kW | 7.0 thru 14.1 | 12.3 thru 17.6 | 7.0 thru 14.1 | 12.3 thru 17.6 |
| Shipping weight - lbs. (kg) 1 package | | | 157 (71) | 165 (75) | 170 (77) | 175 (79) |
| Electrical characteristics | | | 120 volts - 60 hertz - 1 phase | | | |
| **OPTIONAL ACCESSORIES (MUST BE ORDERED EXTRA)** | | | | | | |
| Auxiliary Pump | | | **53J76** - 1 in. NPT | **53J76** - 1 in. NPT | **53J76** - 1 in. NPT | **53J76** - 1 in. NPT |
| Anti-Thermal Siphon Kit | | | **73J84** | **73J84** | **73J84** | **73J84** |
| Downflow/Horizontal Kit | | | **74J25** | **74J25** | **74J25** | **74J25** |
| Horizontal Support Frame Kit | | | **56J18** | **56J18** | **56J18** | **56J18** |
| Twinning Kit (Requires Heating Zone Control Board, see HM30 Optional Accessories) | | | **15L38** | **15L38** | **15L38** | **15L38** |

†NOTE - If AM30Q3/4-90, AM30Q4/5-90 or AM30Q3/4-105 Air Module is matched with HM30-100 Heat Module, maximum output is only 90% of rated HM30 heating capacity.
*At 140°F (60°C) entering water temperature, 65°F (18°C) entering air temperature.
★Polyurethane frame type filter.
†140°F (60°C) water temperature.

## ELECTRICAL DATA

| Unit | Minimum Circuit Ampacity |
|---|---|
| AM30Q2-40 | 5.6 |
| AM30Q2/3-70 | 8.6 |
| AM30Q3/4-70 | 12.2 |
| AM30Q3/4-90 | 12.4 |
| AM30Q3/4-105 | 12.4 |
| AM30Q4/5-90 | 13.8 |
| AM30Q4/5-120 | 13.8 |
| HM30-100 | 5.0 |
| HM30-150 | 5.0 |

| Unit | Minimum Circuit Ampacity |
|---|---|
| AM30Q2-40 with HM30 | 6.9 |
| AM30Q2/3-70 with HM30 | 9.9 |
| AM30Q3/4-70 with HM30 | 13.5 |
| AM30Q3/4-90 with HM30 | 13.7 |
| AM30Q3/4-105 with HM30 | 13.7 |
| AM30Q4/5-90 with HM30 | 15.0 |
| AM30Q4/5-120 with HM30 | 15.0 |

| Unit | Maximum Fuse or Circuit Breaker Size |
|---|---|
| AM30 only | 15.0 |
| HM30 only | 15.0 |
| AM30 with HM30 | 15.0 |

* Where current does not exceed 100 amps, HACR type circuit breaker may be used in place of fuse (U.S. only)

## HEATING EFFICIENCY AND SYSTEM MATCH-UPS

| Heat Module Model No. | Air Module Model No. | ●CAE Combined Annual Efficiency | *$CA_{afue}$ Effective Space Heating Efficiency |
|---|---|---|---|
| **HM30-100** | AM30Q2-40 | .90 | .90 |
| | AM30Q2/3-70 / AM30Q3/4-70 | | |
| | AM30Q3/4-90 / AM30Q4/5-90 | | |
| | AM30Q3/4-105 | | |
| **HM30-150** | AM30Q2-40 | .90 | .90 |
| | AM30Q2/3-70 / AM30Q3/4-70 | | |
| | AM30Q3/4-90 / AM30Q4/5-90 | | |
| | AM30Q3/4-105 | | |
| | AM30Q4/5-120 | | |

NOTE - Tested in accordance with ANSI/ASHRAE Standard 124 “Methods of Testing for Rating Combination Space-Heating and Water Heating Appliances”.
●CAE (Combined Annual Efficiency) = The overall efficiency of the appliance in providing both space heating and water heating.
*$CA_{afue}$ (Effective Space Heating Efficiency) = The effective efficiency of the combined appliance in performing the function of space heating.
**NOTE - $CA_{afue}$ is the same rating as the Annual Fuel Utilization Efficiency (A.F.U.E.) as determined by U.S. Department of Energy test procedures.**

## INTAKE AND EXHAUST PIPE VENTING TABLES

| Vent Pipe Maximum Equivalent Length | | Minimum Pipe Diameter HM30-100 | | Minimum Pipe Diameter HM30-150 | |
|---|---|---|---|---|---|
| **Feet** | **Meters** | **in.** | **mm** | **in.** | **mm** |
| 20 | 6.1 | 2 | 51 | 2 | 51 |
| 30 | 9.1 | 2 or †**3 | 51 or †**76 | 2 or †**3 | 51 or †**76 |
| 40 | 12.2 | 2 or †**3 | 51 or †**76 | 2 or †**3 | 51 or †**76 |
| 50 | 15.2 | 2 or †**3 | 51 or †**76 | **3 | **76 |
| 60 | 18.3 | **3 | **76 | **3 | **76 |
| 70 | 21.3 | 3 | 76 | 3 | 76 |
| 80 | 24.4 | 3 | 76 | 3 | 76 |
| 90 | 27.4 | 3 | 76 | 3 | 76 |
| 100 | 30.5 | 3 | 76 | 3 | 76 |
| 110 | 33.5 | 3 | 76 | 3 | 76 |
| 120 | 36.6 | 3 | 76 | 3 | 76 |
| 130 | 36.6 | 3 | 76 | 3 | 76 |

†NOTE - Use of larger diameter pipe will ensure maximum heat input.
**NOTE - Exhaust pipe requires the use of the 2 inch (51 mm) termination kit instead of 3 inch (76 mm) kit.

**VENTING NOTES -**

One 90° elbow is equivalent to 5 feet (1.5 m) of straight vent pipe.
Two 45° elbows are equal to one 90° elbow.
One 45° elbow is equivalent to 2.5 feet (.75 m) of straight vent pipe.
Intake and Exhaust pipes must be the same diameter.
Exhaust pipe must terminate at 1-1/2 inch (38 mm) diameter pipe for HM30 using 2 inch (51 mm) diameter pipe runs and 2 inch (51 mm) termination kits.
Exhaust pipe must terminate at 2 inch (51 mm) diameter pipe for HM30 using 3 inch (76 mm) diameter pipe runs and 3 inch (76 mm) termination kits.
See pages 22 thru 24 for Optional Termination Kits available.

MINIMUM PIPE LENGTH FOR HM30 - 1 foot (305 mm) with one 90° elbow and 2 inch (51 mm) diameter pipe. 6 equivalent feet (1.8 m) total.

## HIGH ALTITUDE INFORMATION

No derate is required for HM30 units at altitudes up to 7500 feet (2286 m).

# BLOWER DATA

NOTE - All air data is measured external to unit and includes hot water coil air resistance with air filter in place.

### AM30Q2-40 BLOWER PERFORMANCE

| External Static Pressure | | Air Volume at Specific Blower Taps | | | | | |
|---|---|---|---|---|---|---|---|
| | | High | | Medium | | Low | |
| in. w.g. | Pa | cfm | L/s | cfm | L/s | cfm | L/s |
| 0 | 0 | 1100 | 520 | 850 | 400 | 680 | 320 |
| .10 | 25 | 1085 | 510 | 830 | 390 | 670 | 315 |
| .20 | 50 | 1045 | 495 | 800 | 380 | 640 | 300 |
| .30 | 75 | 1005 | 475 | 770 | 365 | 610 | 290 |
| .40 | 100 | 950 | 450 | 730 | 345 | 580 | 275 |
| .50 | 125 | 890 | 420 | 690 | 325 | 530 | 250 |
| .60 | 150 | 810 | 380 | 630 | 295 | 480 | 225 |
| .70 | 175 | 730 | 345 | 570 | 270 | 410 | 195 |
| .80 | 200 | 640 | 300 | 490 | 230 | 330 | 155 |
| .90 | 225 | 520 | 245 | 380 | 180 | - - - - | - - - - |

### AM30Q2/3-70 BLOWER PERFORMANCE

| External Static Pressure | | Air Volume at Specific Blower Taps | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | High | | Medium-High | | Medium-Low | | Low | |
| in. w.g. | Pa | cfm | L/s | cfm | L/s | cfm | L/s | cfm | L/s |
| 0 | 0 | 1500 | 710 | 1310 | 620 | 1070 | 505 | 840 | 395 |
| .10 | 25 | 1440 | 680 | 1270 | 600 | 1050 | 495 | 820 | 385 |
| .20 | 50 | 1370 | 645 | 1220 | 575 | 1020 | 480 | 800 | 380 |
| .30 | 75 | 1300 | 615 | 1180 | 555 | 980 | 460 | 770 | 365 |
| .40 | 100 | 1240 | 585 | 1120 | 530 | 930 | 440 | 740 | 350 |
| .50 | 125 | 1170 | 550 | 1050 | 495 | 870 | 410 | 700 | 330 |
| .60 | 150 | 1090 | 515 | 960 | 455 | 800 | 380 | 650 | 305 |
| .70 | 175 | 990 | 465 | 870 | 410 | 720 | 340 | 590 | 380 |
| .80 | 200 | 880 | 415 | 770 | 365 | 630 | 295 | 500 | 235 |
| .90 | 225 | 750 | 355 | 650 | 305 | 520 | 245 | - - - - | - - - - |

### AM30Q3/4-70 & AM30Q3/4-90 BLOWER PERFORMANCE

| External Static Pressure | | Air Volume at Specific Blower Taps | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | High | | Medium-High | | Medium-Low | | Low | |
| in. w.g. | Pa | cfm | L/s | cfm | L/s | cfm | L/s | cfm | L/s |
| 0 | 0 | 2060 | 970 | 1640 | 775 | 1420 | 670 | 1110 | 525 |
| .10 | 25 | 2000 | 945 | 1640 | 775 | 1430 | 675 | 1110 | 525 |
| .20 | 50 | 1930 | 910 | 1630 | 770 | 1420 | 670 | 1120 | 530 |
| .30 | 75 | 1850 | 875 | 1600 | 755 | 1410 | 665 | 1110 | 525 |
| .40 | 100 | 1770 | 835 | 1550 | 730 | 1370 | 645 | 1090 | 515 |
| .50 | 125 | 1670 | 790 | 1480 | 700 | 1300 | 615 | 1060 | 500 |
| .60 | 150 | 1560 | 735 | 1390 | 655 | 1230 | 580 | 1010 | 475 |
| .70 | 175 | 1420 | 670 | 1280 | 605 | 1130 | 535 | 930 | 440 |
| .80 | 200 | 1280 | 605 | 1140 | 540 | 1020 | 480 | 830 | 390 |
| .90 | 225 | 1100 | 520 | 980 | 460 | 880 | 415 | 700 | 330 |

### AM30Q4/5-90 BLOWER PERFORMANCE

| External Static Pressure | | Air Volume at Specific Blower Taps | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | High | | Medium-High | | Medium | | Medium-Low | | Low | |
| in. w.g. | Pa | cfm | L/s | cfm | L/s | cfm | L/s | cfm | L/s | cfm | L/s |
| 0 | 0 | 2350 | 1110 | 2020 | 955 | 1830 | 865 | 1630 | 770 | 1430 | 675 |
| .10 | 25 | 2330 | 1100 | 2020 | 955 | 1820 | 860 | 1620 | 765 | 1430 | 675 |
| .20 | 50 | 2300 | 1085 | 2010 | 950 | 1810 | 855 | 1610 | 760 | 1420 | 670 |
| .30 | 75 | 2260 | 1065 | 1990 | 940 | 1780 | 840 | 1590 | 750 | 1410 | 665 |
| .40 | 100 | 2200 | 1040 | 1950 | 920 | 1750 | 825 | 1560 | 735 | 1400 | 660 |
| .50 | 125 | 2140 | 1010 | 1900 | 895 | 1720 | 810 | 1530 | 720 | 1370 | 645 |
| .60 | 150 | 2060 | 970 | 1860 | 875 | 1680 | 795 | 1500 | 710 | 1340 | 630 |
| .70 | 175 | 1970 | 930 | 1780 | 840 | 1610 | 760 | 1460 | 690 | 1300 | 615 |
| .80 | 200 | 1890 | 890 | 1700 | 800 | 1550 | 730 | 1400 | 660 | 1240 | 585 |
| .90 | 225 | 1780 | 840 | 1610 | 760 | 1480 | 700 | 1330 | 630 | 1180 | 555 |

### AM30Q3/4-105 BLOWER PERFORMANCE

| External Static Pressure | | Air Volume at Specific Blower Taps | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | High | | Medium-High | | Medium-Low | | Low | |
| in. w.g. | Pa | cfm | L/s | cfm | L/s | cfm | L/s | cfm | L/s |
| 0 | 0 | 2000 | 945 | 1640 | 775 | 1430 | 675 | 1110 | 525 |
| .10 | 25 | 1930 | 910 | 1630 | 770 | 1430 | 675 | 1110 | 525 |
| .20 | 50 | 1870 | 880 | 1610 | 760 | 1410 | 665 | 1110 | 525 |
| .30 | 75 | 1790 | 845 | 1570 | 740 | 1380 | 650 | 1100 | 520 |
| .40 | 100 | 1700 | 800 | 1500 | 710 | 1340 | 630 | 1080 | 510 |
| .50 | 125 | 1600 | 755 | 1430 | 675 | 1260 | 595 | 1050 | 495 |
| .60 | 150 | 1480 | 700 | 1340 | 630 | 1180 | 555 | 980 | 460 |
| .70 | 175 | 1350 | 635 | 1200 | 565 | 1090 | 515 | 890 | 420 |
| .80 | 200 | 1200 | 565 | 1060 | 500 | 970 | 460 | 780 | 370 |
| .90 | 225 | 1050 | 495 | 900 | 425 | 810 | 380 | 650 | 305 |

### AM30Q4/5-120 BLOWER PERFORMANCE

| External Static Pressure | | Air Volume at Specific Blower Taps | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | High | | Medium-High | | Medium | | Medium-Low | | Low | |
| in. w.g. | Pa | cfm | L/s | cfm | L/s | cfm | L/s | cfm | L/s | cfm | L/s |
| 0 | 0 | 2400 | 1135 | 2100 | 990 | 1890 | 890 | 1700 | 800 | 1500 | 710 |
| .10 | 25 | 2350 | 1110 | 2080 | 980 | 1880 | 885 | 1690 | 800 | 1500 | 710 |
| .20 | 50 | 2300 | 1085 | 2050 | 965 | 1860 | 880 | 1680 | 795 | 1490 | 705 |
| .30 | 75 | 2230 | 1050 | 2010 | 950 | 1830 | 865 | 1650 | 780 | 1470 | 695 |
| .40 | 100 | 2160 | 1020 | 1970 | 930 | 1800 | 850 | 1620 | 765 | 1440 | 680 |
| .50 | 125 | 2090 | 985 | 1920 | 905 | 1750 | 825 | 1590 | 750 | 1410 | 665 |
| .60 | 150 | 2010 | 950 | 1860 | 880 | 1700 | 800 | 1540 | 725 | 1380 | 650 |
| .70 | 175 | 1930 | 910 | 1800 | 850 | 1640 | 775 | 1500 | 710 | 1330 | 630 |
| .80 | 200 | 1840 | 870 | 1710 | 805 | 1580 | 745 | 1430 | 675 | 1290 | 610 |
| .90 | 225 | 1750 | 825 | 1650 | 780 | 1500 | 710 | 1350 | 635 | 1230 | 580 |

# HEATING PERFORMANCE

## AM30Q2-40 HEATING OUTPUTS

| Air Volume | | Heating Outputs At Various Water Temperatures | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 120°F (49°C) | | 130°F (54°C) | | 140°F (60°C) | | 150°F (66°C) | | 160°F (71°C) | | 170°F (77°C) | |
| cfm | L/s | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW |
| 300 | 140 | 13,800 | 4.0 | 16,300 | 4.8 | 18,800 | 5.5 | 21,300 | 6.2 | 23,800 | 7.0 | 26,300 | 7.7 |
| 400 | 190 | 16,800 | 4.9 | 19,900 | 5.8 | 23,000 | 6.7 | 26,000 | 7.6 | 29,100 | 8.5 | 32,100 | 9.4 |
| 500 | 235 | 19,500 | 5.7 | 23,100 | 6.8 | 26,600 | 7.8 | 30,200 | 8.8 | 33,700 | 9.9 | 37,300 | 10.9 |
| 600 | 285 | 21,900 | 6.4 | 25,900 | 7.6 | 29,800 | 8.7 | 33,800 | 9.9 | 37,800 | 11.7 | 41,800 | 12.2 |
| 700 | 330 | 24,000 | 7.0 | 28,400 | 8.3 | 32,700 | 9.6 | 37,100 | 10.9 | 41,400 | 12.1 | 45,800 | 13.4 |
| 800 | 380 | 25,900 | 7.6 | 30,600 | 9.0 | 35,300 | 10.3 | 40,000 | 11.7 | 44,700 | 13.1 | 49,400 | 14.5 |
| 900 | 425 | 27,600 | 8.1 | 32,700 | 9.6 | 37,700 | 11.0 | 42,700 | 12.5 | 47,800 | 14.0 | 52,800 | 15.5 |
| 1000 | 470 | 29,300 | 8.6 | 34,600 | 10.1 | 39,900 | 11.7 | 45,200 | 13.2 | 50,500 | 14.8 | 55,800 | 16.3 |
| 1100 | 520 | 30,700 | 9.0 | 36,300 | 10.6 | 41,900 | 12.3 | 47,500 | 13.9 | 53,100 | 15.6 | 58,700 | 17.2 |
| 1200 | 565 | 32,100 | 9.4 | 37,900 | 11.1 | 43,800 | 12.8 | 49,600 | 14.5 | 55,500 | 16.3 | 61,300 | 18.0 |

## AM30Q2/3-70 HEATING OUTPUTS

| Air Volume | | Heating Outputs At Various Water Temperatures | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 120°F (49°C) | | 130°F (54°C) | | 140°F (60°C) | | 150°F (66°C) | | 160°F (71°C) | | 170°F (77°C) | |
| cfm | L/s | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW |
| 500 | 236 | 26,300 | 7.7 | 31,000 | 9.1 | 35,800 | 10.5 | 40,600 | 11.9 | 45,400 | 13.3 | 50,200 | 14.7 |
| 600 | 283 | 30,200 | 8.8 | 35,700 | 10.5 | 41,100 | 12.0 | 46,600 | 13.7 | 52,100 | 15.3 | 57,600 | 16.9 |
| 700 | 330 | 33,700 | 9.9 | 39,800 | 11.7 | 46,000 | 13.5 | 52,100 | 15.3 | 58,200 | 17.1 | 64,400 | 18.9 |
| 800 | 378 | 37,000 | 10.8 | 43,700 | 12.8 | 50,400 | 14.8 | 57,100 | 16.7 | 63,800 | 18.7 | 70,600 | 20.7 |
| 900 | 425 | 39,900 | 11.7 | 47,200 | 13.8 | 54,400 | 15.9 | 61,700 | 18.1 | 69,000 | 20.2 | 76,200 | 22.3 |
| 1000 | 472 | 42,700 | 12.5 | 50,400 | 14.8 | 58,200 | 17.1 | 66,000 | 19.3 | 73,700 | 21.6 | 81,500 | 23.9 |
| 1100 | 519 | 45,200 | 13.2 | 53,000 | 15.5 | 61,700 | 18.1 | 66,900 | 19.6 | 78,100 | 22.9 | 86,300 | 25.3 |
| 1200 | 566 | 47,600 | 13.9 | 56,200 | 16.5 | 64,900 | 19.0 | 73,500 | 21.5 | 82,200 | 24.1 | 90,800 | 26.6 |
| 1300 | 614 | 49,800 | 14.6 | 58,800 | 17.2 | 67,900 | 19.9 | 76,900 | 22.5 | 86,000 | 25.2 | 95,000 | 27.8 |
| 1400 | 661 | 51,800 | 15.2 | 61,300 | 18.0 | 70,700 | 20.7 | 80,100 | 23.5 | 89,500 | 26.2 | 99,000 | 29.0 |

## AM30Q3/4-70 HEATING OUTPUTS

| Air Volume | | Heating Outputs At Various Water Temperatures | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 120°F (49°C) | | 130°F (54°C) | | 140°F (60°C) | | 150°F (66°C) | | 160°F (71°C) | | 170°F (77°C) | |
| cfm | L/s | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW |
| 650 | 305 | 32,700 | 9.6 | 38,600 | 11.3 | 44,600 | 13.1 | 50,500 | 14.8 | 56,500 | 16.6 | 62,400 | 18.3 |
| 800 | 380 | 37,900 | 11.1 | 44,800 | 13.1 | 51,600 | 15.1 | 58,500 | 17.1 | 65,400 | 19.2 | 72,300 | 21.2 |
| 950 | 450 | 42,400 | 12.4 | 50,200 | 14.7 | 57,900 | 17.0 | 65,600 | 19.2 | 73,300 | 21.5 | 81,000 | 23.7 |
| 1100 | 520 | 46,500 | 13.6 | 55,000 | 16.1 | 63,400 | 18.6 | 71,900 | 21.1 | 80,300 | 23.5 | 88,800 | 26.0 |
| 1250 | 590 | 50,100 | 14.7 | 59,200 | 17.3 | 68,400 | 20.0 | 77,500 | 22.7 | 86,600 | 24.4 | 95,700 | 28.0 |
| 1400 | 660 | 53,400 | 15.6 | 63,100 | 18.5 | 72,800 | 21.3 | 82,500 | 24.2 | 92,300 | 27.0 | 102,000 | 29.9 |
| 1550 | 730 | 56,400 | 16.5 | 66,600 | 19.5 | 76,900 | 22.5 | 87,100 | 25.5 | 97,400 | 28.5 | 107,600 | 31.5 |
| 1700 | 800 | 59,100 | 17.3 | 69,800 | 20.5 | 80,600 | 23.6 | 91,300 | 26.8 | 102,100 | 29.9 | 112,800 | 33.1 |
| 1850 | 875 | 60,600 | 17.8 | 72,800 | 21.3 | 84,000 | 24.6 | 95,200 | 27.9 | 106,400 | 31.2 | 117,600 | 34.5 |
| 2000 | 945 | 63,900 | 18.7 | 75,500 | 22.1 | 87,100 | 25.5 | 98,700 | 28.9 | 110,300 | 32.3 | 122,000 | 35.7 |

## AM30Q3/4-90 HEATING OUTPUTS

| Air Volume | | Heating Outputs At Various Water Temperatures | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 120°F (49°C) | | 130°F (54°C) | | 140°F (60°C) | | 150°F (66°C) | | 160°F (71°C) | | 170°F (77°C) | |
| cfm | L/s | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW |
| 650 | 305 | 34,000 | 10.0 | 40,100 | 11.7 | 46,300 | 13.7 | 52,400 | 15.4 | 58,600 | 17.2 | 64,800 | 19.0 |
| 800 | 380 | 39,900 | 11.7 | 46,900 | 13.7 | 54,200 | 15.9 | 61,400 | 18.0 | 68,600 | 20.1 | 75,800 | 22.2 |
| 950 | 450 | 45,200 | 13.2 | 53,100 | 15.6 | 61,300 | 18.0 | 69,500 | 20.4 | 77,600 | 22.7 | 85,800 | 25.1 |
| 1100 | 520 | 50,000 | 14.7 | 58,700 | 17.2 | 67,800 | 19.9 | 76,800 | 22.5 | 85,800 | 25.1 | 94,900 | 27.8 |
| 1250 | 590 | 54,400 | 15.9 | 63,900 | 18.7 | 73,700 | 21.6 | 83,500 | 24.5 | 93,300 | 27.3 | 103,200 | 30.2 |
| 1400 | 660 | 58,400 | 17.1 | 68,600 | 20.1 | 79,100 | 23.2 | 89,700 | 26.3 | 100,200 | 29.4 | 110,800 | 32.5 |
| 1550 | 730 | 62,200 | 18.2 | 72,900 | 21.4 | 84,100 | 24.6 | 95,400 | 28.0 | 106,600 | 31.2 | 117,800 | 34.5 |
| 1700 | 800 | 65,700 | 19.3 | 77,000 | 22.6 | 88,800 | 26.0 | 100,600 | 29.5 | 112,500 | 33.0 | 124,300 | 36.4 |
| 1850 | 875 | 68,900 | 20.2 | 80,700 | 23.6 | 93,100 | 27.3 | 105,600 | 30.9 | 118,000 | 34.6 | 130,400 | 38.2 |
| 2000 | 945 | 72,000 | 21.1 | 84,200 | 24.7 | 97,200 | 28.5 | 110,100 | 32.3 | 123,100 | 36.1 | 136,100 | 39.9 |

## AM30Q4/5-90 HEATING OUTPUTS

| Air Volume | | Heating Outputs At Various Water Temperatures | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 120°F (49°C) | | 130°F (54°C) | | 140°F (60°C) | | 150°F (66°C) | | 160°F (71°C) | | 170°F (77°C) | |
| cfm | L/s | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW |
| 1100 | 520 | 50,000 | 14.7 | 58,700 | 17.2 | 67,800 | 19.9 | 76,800 | 22.5 | 85,800 | 25.1 | 94,900 | 27.8 |
| 1250 | 590 | 54,400 | 15.9 | 63,900 | 18.7 | 73,700 | 21.6 | 83,500 | 24.5 | 93,300 | 27.3 | 103,200 | 30.2 |
| 1400 | 660 | 58,400 | 17.1 | 68,600 | 20.1 | 79,100 | 23.2 | 89,700 | 26.3 | 100,200 | 29.4 | 110,800 | 32.5 |
| 1550 | 730 | 62,200 | 18.2 | 72,900 | 21.4 | 84,100 | 24.6 | 95,400 | 28.0 | 106,600 | 31.2 | 117,800 | 34.5 |
| 1700 | 800 | 65,700 | 19.3 | 77,000 | 22.6 | 88,800 | 26.0 | 100,600 | 29.5 | 112,500 | 33.0 | 124,300 | 36.4 |
| 1850 | 875 | 68,900 | 20.2 | 80,700 | 23.6 | 93,100 | 27.3 | 105,600 | 30.9 | 118,000 | 34.6 | 130,400 | 38.2 |
| 2000 | 945 | 71,200 | 20.9 | 84,200 | 24.7 | 97,200 | 28.5 | 110,100 | 32.3 | 123,100 | 36.1 | 136,100 | 40.0 |
| 2150 | 1015 | 74,100 | 21.7 | 87,500 | 25.6 | 101,100 | 29.6 | 114,400 | 33.5 | 127,900 | 37.5 | 141,400 | 41.4 |
| 2300 | 1085 | 76,700 | 22.5 | 90,600 | 26.5 | 104,500 | 30.6 | 118,500 | 34.7 | 132,400 | 38.8 | 146,400 | 42.9 |
| 2450 | 1155 | 79,100 | 23.2 | 93,500 | 27.4 | 107,900 | 31.6 | 122,300 | 35.8 | 136,700 | 40.1 | 151,100 | 44.3 |

## HEATING PERFORMANCE

### AM30Q3/4-105 HEATING OUTPUTS

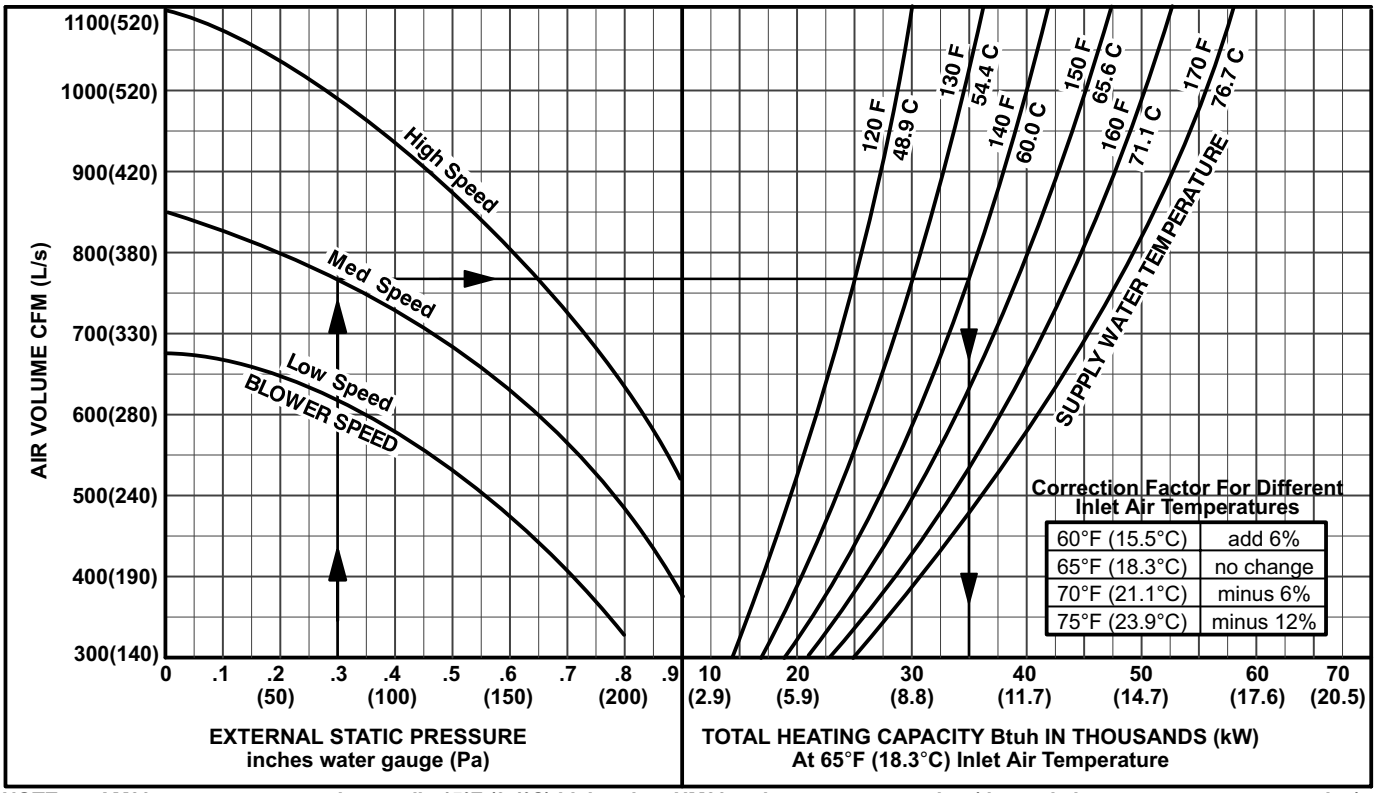
| Air Volume | | Heating Outputs At Various Water Temperatures | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 120°F (49°C) | | 130°F (54°C) | | 140°F (60°C) | | 150°F (66°C) | | 160°F (71°C) | | 170°F (77°C) | |
| cfm | L/s | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW |
| 650 | 305 | 37,500 | 11.0 | 44,400 | 13.0 | 51,200 | 15.0 | 58,000 | 17.0 | 64,900 | 19.0 | 71,700 | 21.0 |
| 800 | 380 | 44,700 | 13.1 | 52,900 | 15.5 | 61,000 | 17.9 | 66,200 | 19.4 | 77,300 | 22.6 | 85,400 | 25.0 |
| 950 | 450 | 51,400 | 15.1 | 60,700 | 17.8 | 70,000 | 20.5 | 79,400 | 23.3 | 88,700 | 26.0 | 98,100 | 28.7 |
| 1100 | 520 | 57,300 | 16.8 | 67,700 | 19.8 | 78,100 | 22.9 | 88,500 | 25.9 | 99,000 | 29.0 | 109,400 | 32.1 |
| 1250 | 590 | 62,900 | 18.4 | 74,300 | 21.8 | 85,700 | 25.1 | 97,200 | 28.5 | 108,600 | 31.8 | 120,000 | 35.2 |
| 1400 | 660 | 68,000 | 19.9 | 80,400 | 23.6 | 92,700 | 27.2 | 105,100 | 30.8 | 117,500 | 34.4 | 129,900 | 38.1 |
| 1550 | 730 | 72,800 | 21.3 | 86,000 | 25.2 | 99,300 | 29.1 | 112,500 | 33.0 | 125,700 | 36.8 | 139,000 | 40.7 |
| 1700 | 800 | 77,200 | 22.6 | 91,200 | 26.7 | 105,300 | 30.9 | 119,300 | 35.0 | 133,400 | 39.1 | 147,400 | 43.2 |
| 1850 | 875 | 81,300 | 23.8 | 96,100 | 28.2 | 110,900 | 32.5 | 125,700 | 36.8 | 140,500 | 41.1 | 155,300 | 15.5 |
| 2000 | 945 | 85,200 | 25.0 | 100,700 | 29.5 | 116,100 | 34.0 | 131,600 | 38.6 | 147,100 | 43.1 | 162,600 | 47.6 |

### AM30Q4/5-120 HEATING OUTPUTS

| Air Volume | | Heating Outputs At Various Water Temperatures | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 120°F (49°C) | | 130°F (54°C) | | 140°F (60°C) | | 150°F (66°C) | | 160°F (71°C) | | 170°F (77°C) | |
| cfm | L/s | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW | Btuh | kW |
| 1100 | 520 | 57,300 | 16.8 | 67,700 | 19.8 | 78,100 | 22.9 | 88,500 | 25.9 | 99,000 | 29.0 | 109,400 | 32.1 |
| 1250 | 590 | 62,900 | 18.4 | 74,300 | 21.8 | 85,700 | 25.1 | 97,200 | 28.4 | 108,600 | 31.8 | 120,000 | 35.2 |
| 1400 | 660 | 68,000 | 19.9 | 80,400 | 23.6 | 92,700 | 27.2 | 105,100 | 30.8 | 117,500 | 34.4 | 129,900 | 38.1 |
| 1550 | 730 | 72,800 | 21.3 | 86,000 | 25.2 | 99,300 | 29.1 | 112,500 | 33.0 | 125,700 | 36.8 | 139,000 | 40.7 |
| 1700 | 800 | 77,200 | 22.6 | 91,200 | 26.7 | 105,300 | 30.9 | 119,300 | 35.0 | 133,400 | 39.1 | 147,400 | 43.2 |
| 1850 | 875 | 81,300 | 23.8 | 96,100 | 28.2 | 110,900 | 32.5 | 125,700 | 36.8 | 140,500 | 41.2 | 155,300 | 45.5 |
| 2000 | 945 | 85,200 | 25.0 | 100,700 | 29.5 | 116,100 | 34.0 | 131,600 | 38.6 | 147,100 | 43.1 | 162,600 | 47.6 |
| 2150 | 1015 | 88,700 | 26.0 | 104,900 | 30.7 | 121,000 | 35.5 | 137,200 | 40.2 | 153,300 | 44.9 | 169,400 | 49.6 |
| 2300 | 1085 | 92,100 | 27.0 | 108,900 | 31.9 | 125,600 | 36.8 | 142,400 | 41.7 | 159,100 | 46.6 | 175,900 | 51.5 |
| 2450 | 1155 | 95,300 | 27.9 | 112,600 | 33.0 | 129,900 | 38.1 | 147,200 | 43.1 | 164,600 | 48.2 | 181,900 | 53.3 |

## HEATING PERFORMANCE CURVES

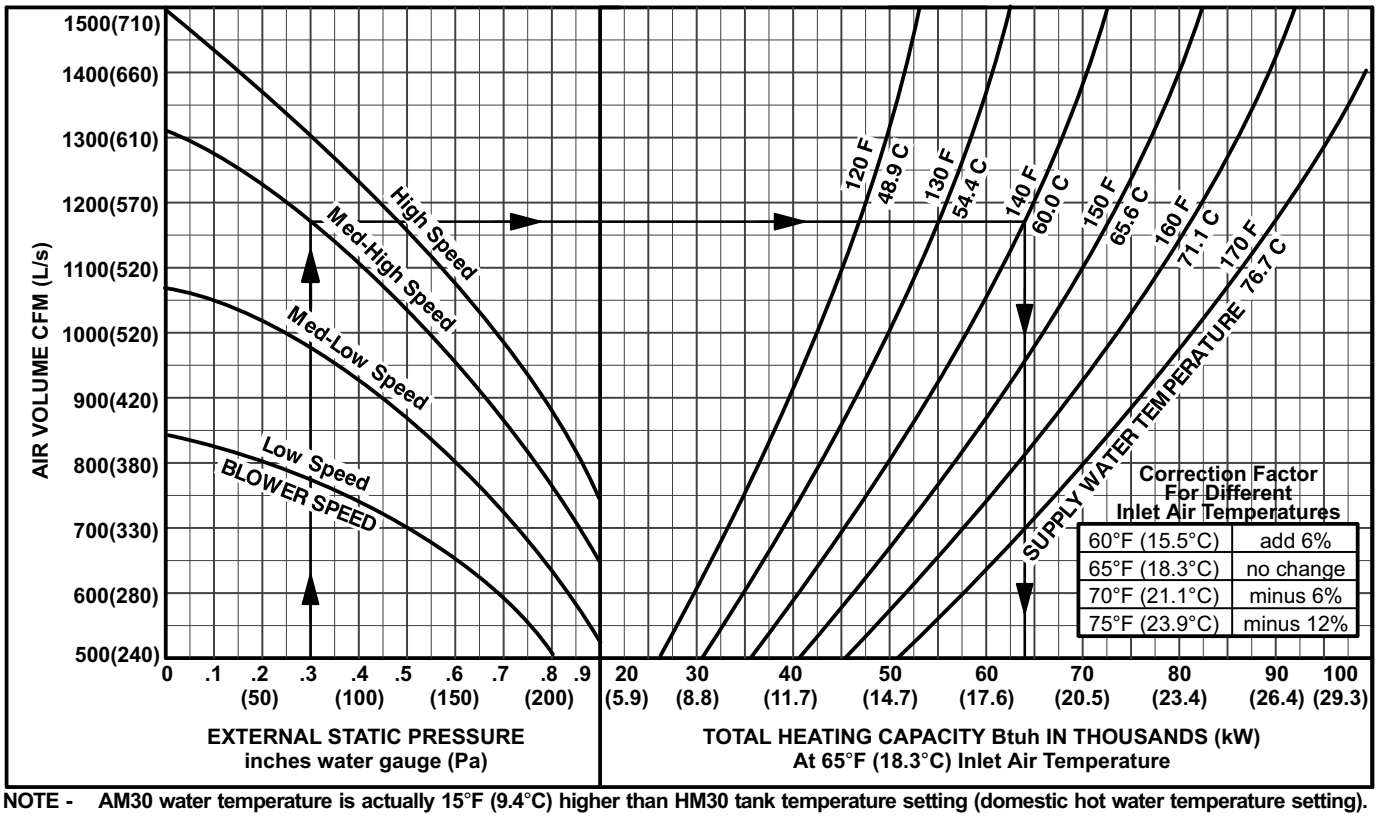
### AM30Q2-40 HEATING PERFORMANCE CURVE




**NOTE - AM30 water temperature is actually 15°F (9.4°C) higher than HM30 tank temperature setting (domestic hot water temperature setting).**

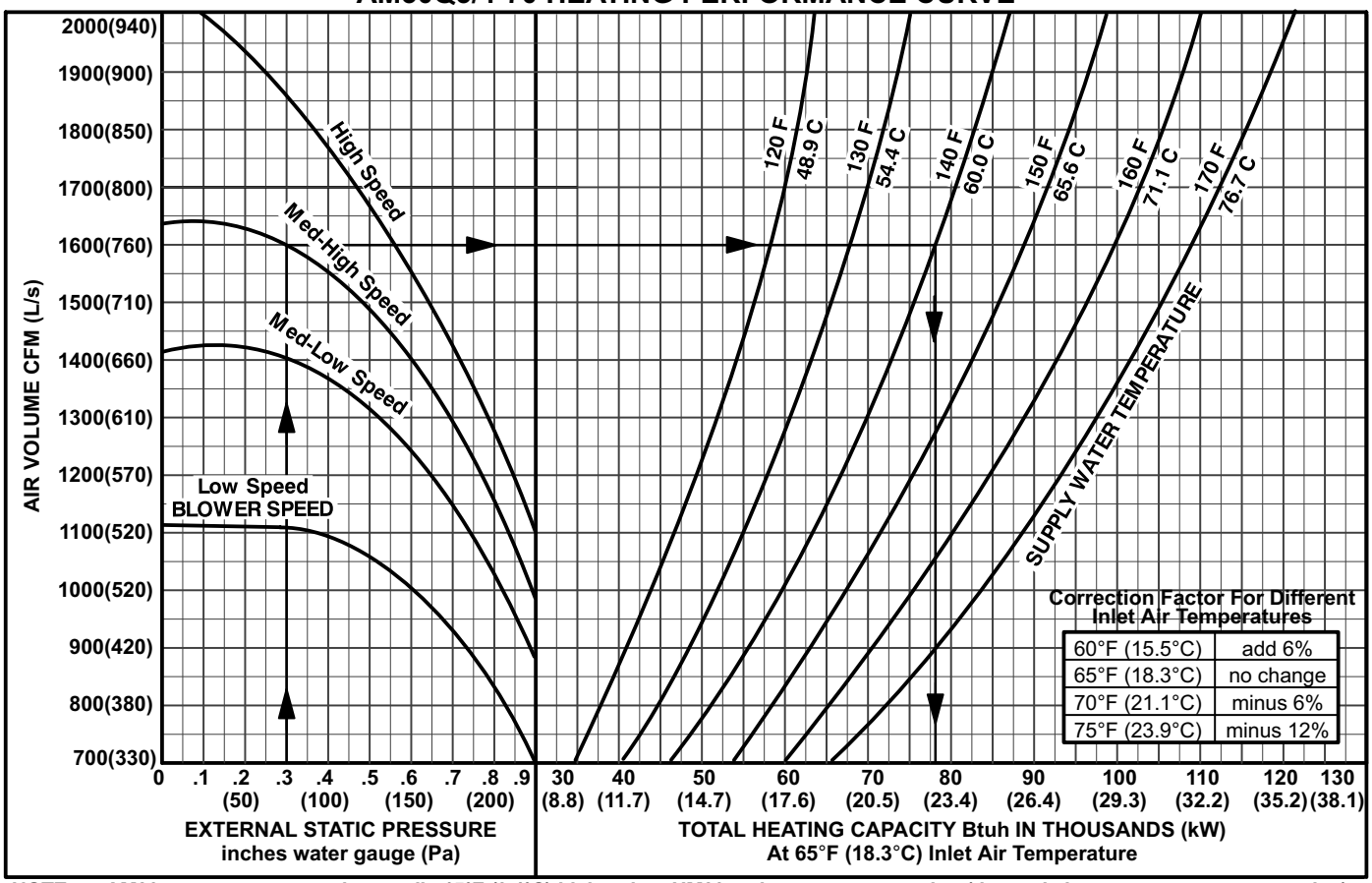
## HEATING PERFORMANCE CURVES

### AM30Q2/3-70 HEATING PERFORMANCE CURVE




**NOTE - AM30 water temperature is actually 15°F (9.4°C) higher than HM30 tank temperature setting (domestic hot water temperature setting).**

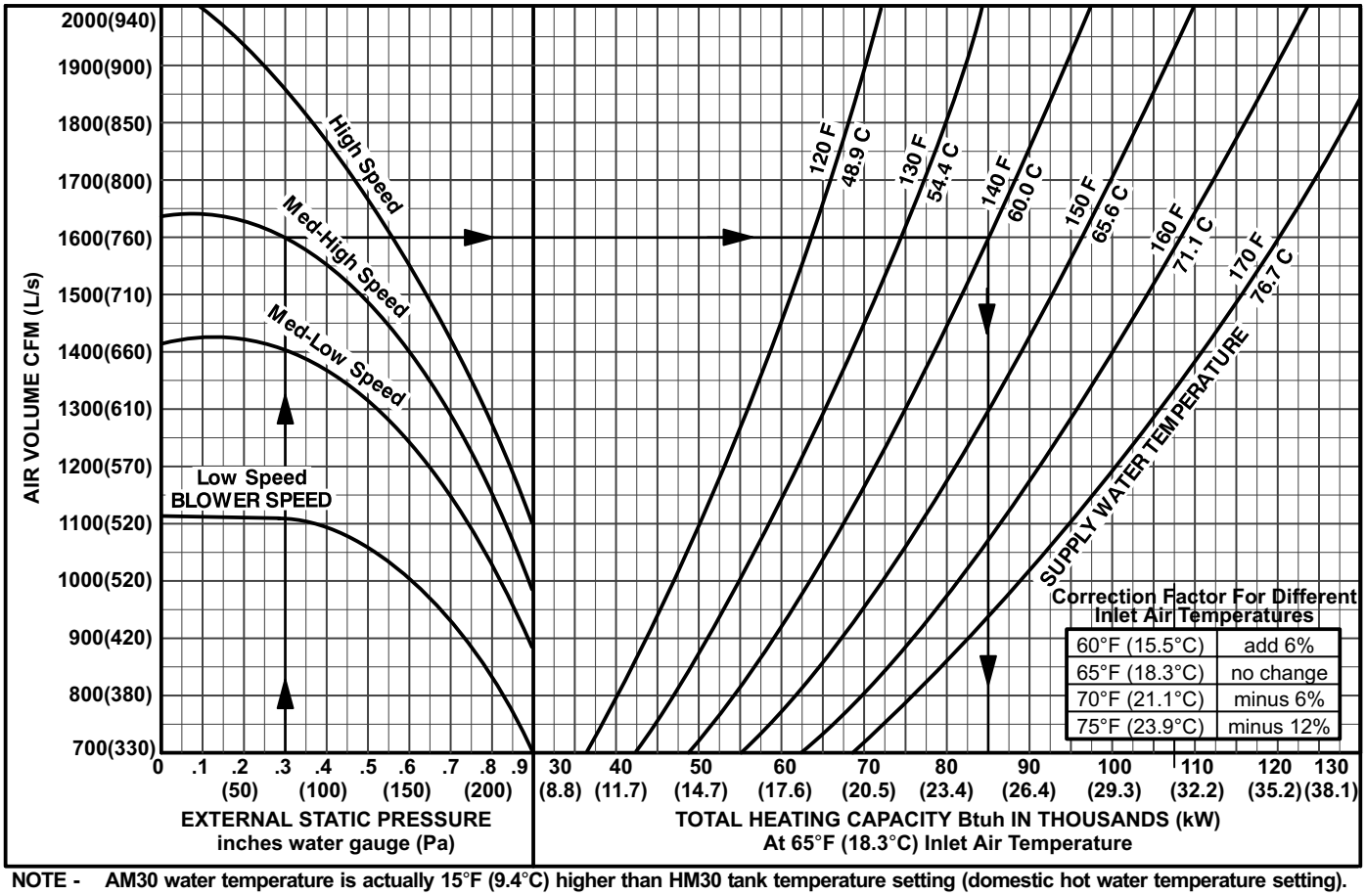
### AM30Q3/4-70 HEATING PERFORMANCE CURVE




**NOTE - AM30 water temperature is actually 15°F (9.4°C) higher than HM30 tank temperature setting (domestic hot water temperature setting).**

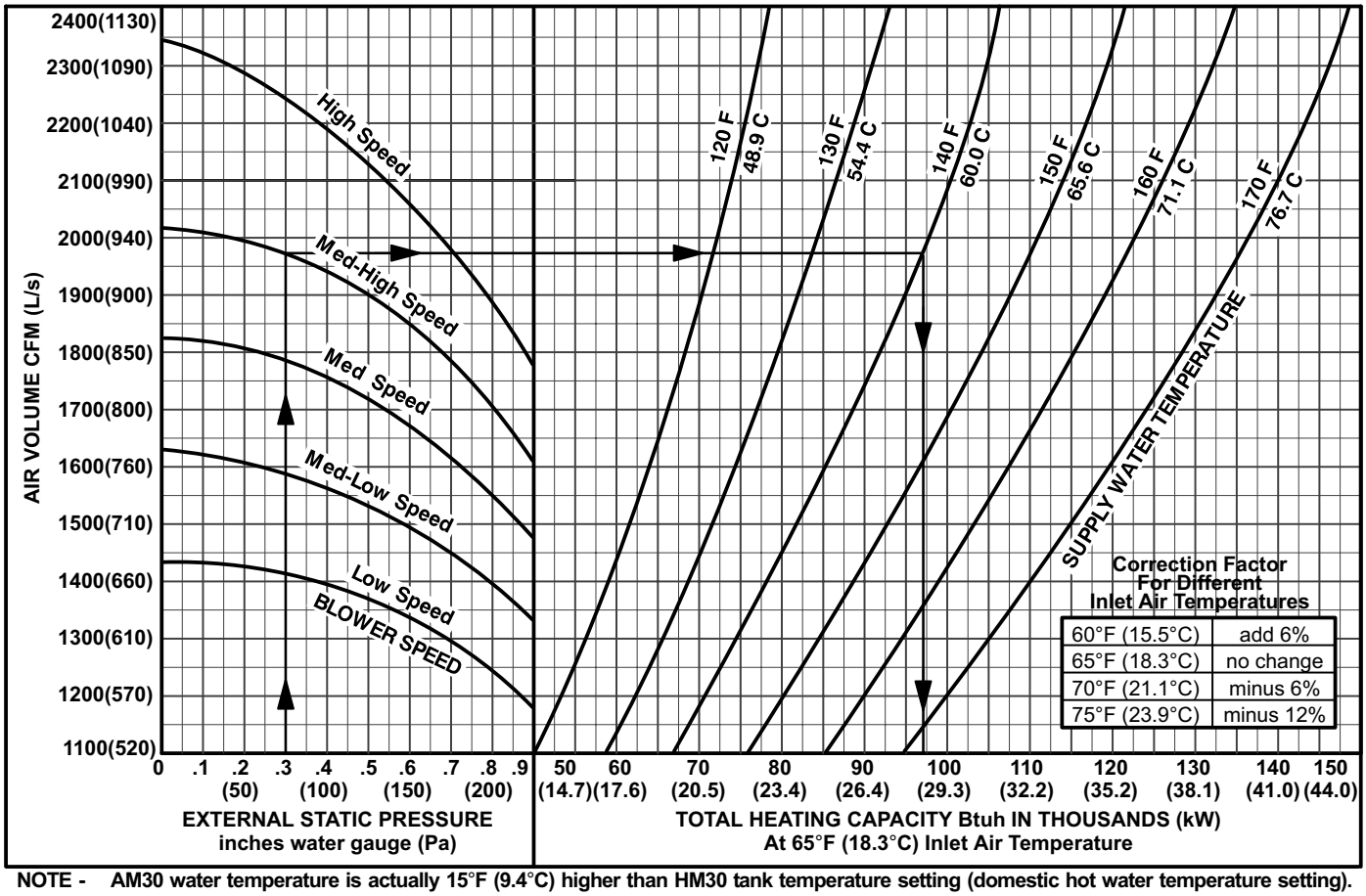
## HEATING PERFORMANCE CURVES

### AM30Q3/4-90 HEATING PERFORMANCE CURVE




**NOTE - AM30 water temperature is actually 15°F (9.4°C) higher than HM30 tank temperature setting (domestic hot water temperature setting).**

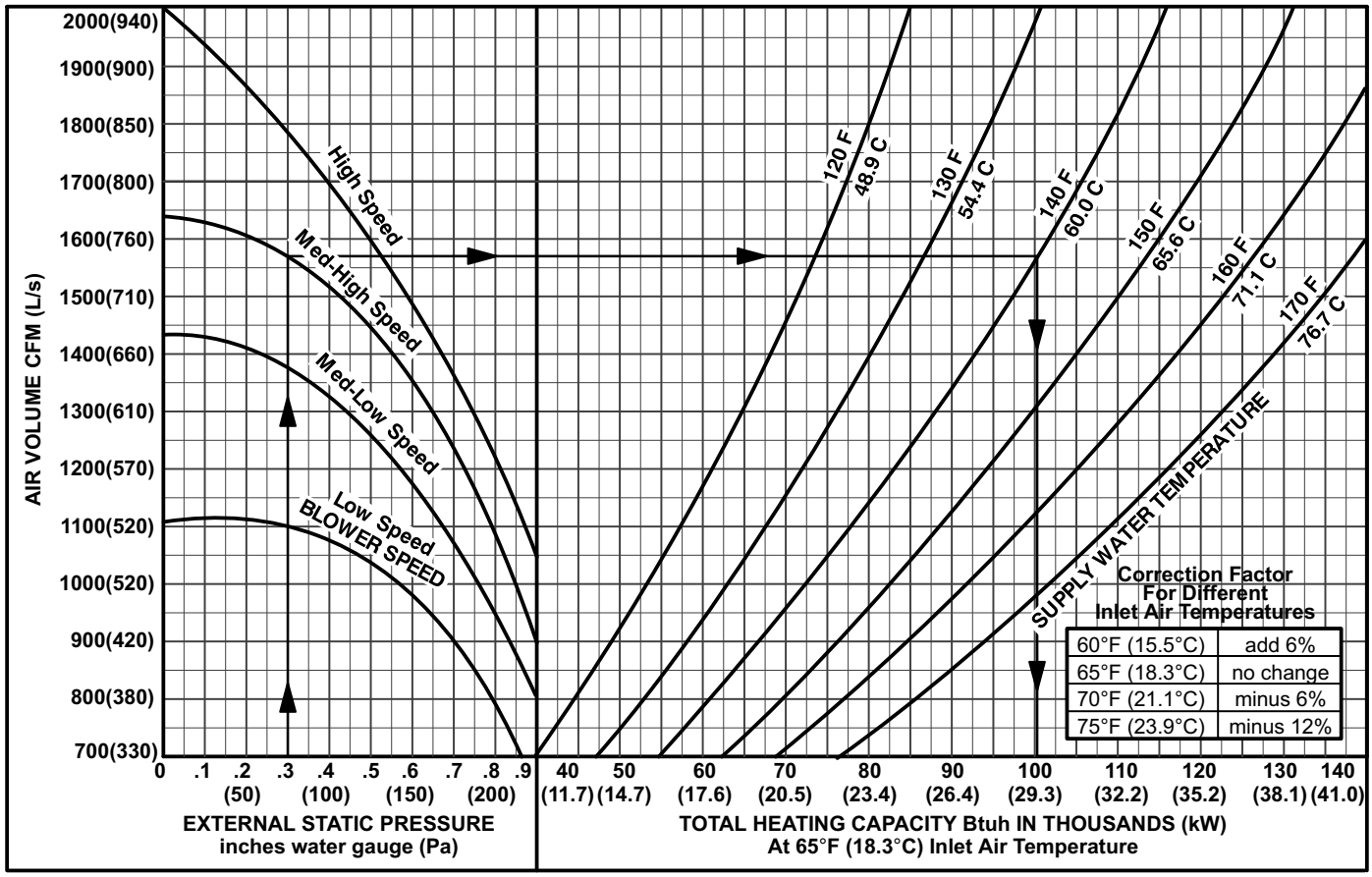
### AM30Q4/5-90 HEATING PERFORMANCE CURVE




**NOTE - AM30 water temperature is actually 15°F (9.4°C) higher than HM30 tank temperature setting (domestic hot water temperature setting).**

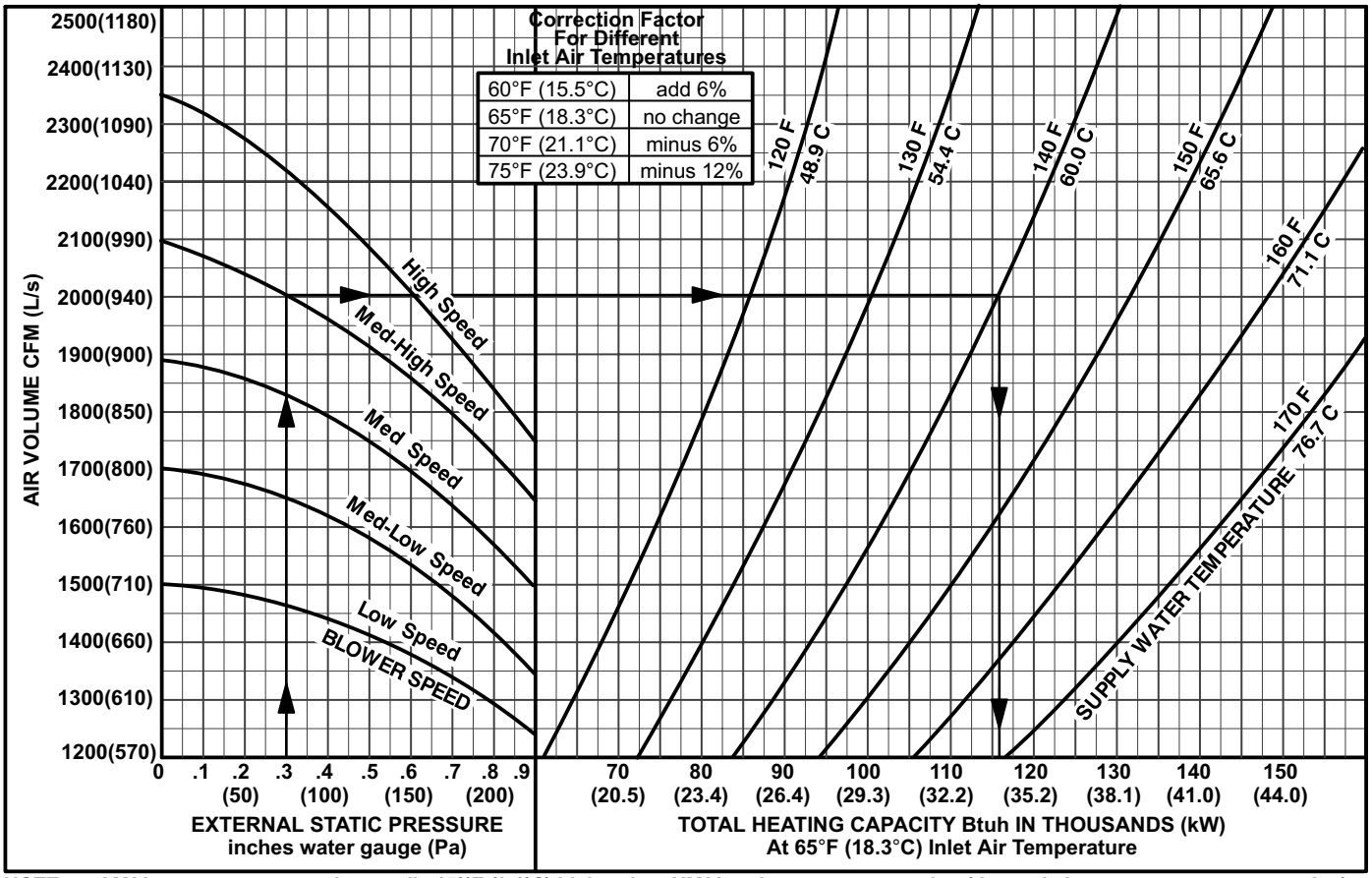
## HEATING PERFORMANCE CURVES

### AM30Q3/4-105 HEATING PERFORMANCE CURVE




**NOTE - AM30 water temperature is actually 15°F (9.4°C) higher than HM30 tank temperature setting (domestic hot water temperature setting).**

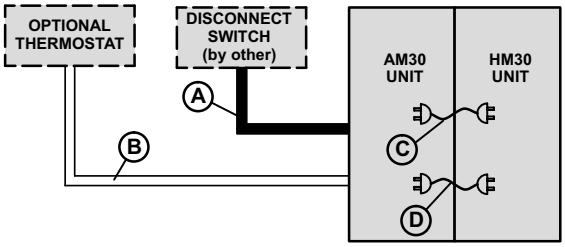
### AM30Q4/5-120 HEATING PERFORMANCE CURVE




**NOTE - AM30 water temperature is actually 15°°F (9.4°C) higher than HM30 tank temperature setting (domestic hot water temperature setting).**

# FIELD WIRING

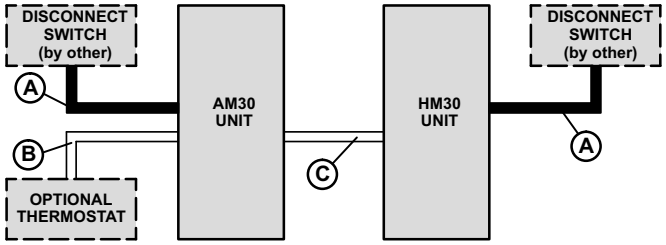
### CLOSE-COUPLED APPLICATION




A - Two wire power with ground
B - Three wire control voltage (Heating Only)
Four wire control voltage (Cooling)
C - Four wire control voltage (Harness furnished with AM30)
D - Two wire power (Harness furnished with AM30)
*- Field wiring not furnished -*

All wiring must conform to NEC or CEC and local electrical codes.

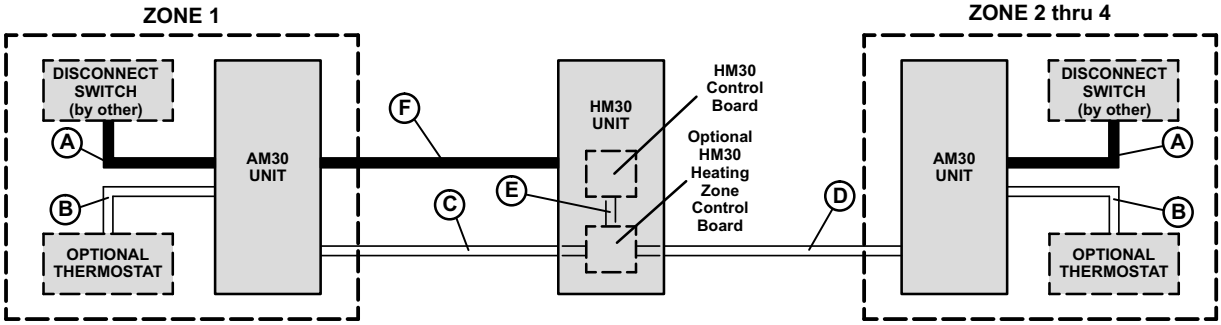
### REMOTE APPLICATION




A - Two wire power with ground
B - Three wire control voltage (Heating Only)
Four wire control voltage (Cooling)
C - Four wire control voltage
*- Field wiring not furnished -*

All wiring must conform to NEC or CEC and local electrical codes.

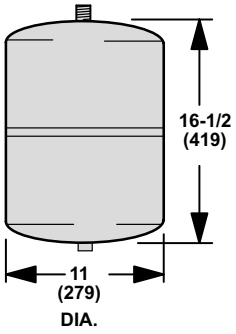
### ZONING APPLICATION (Up To Four Zones)



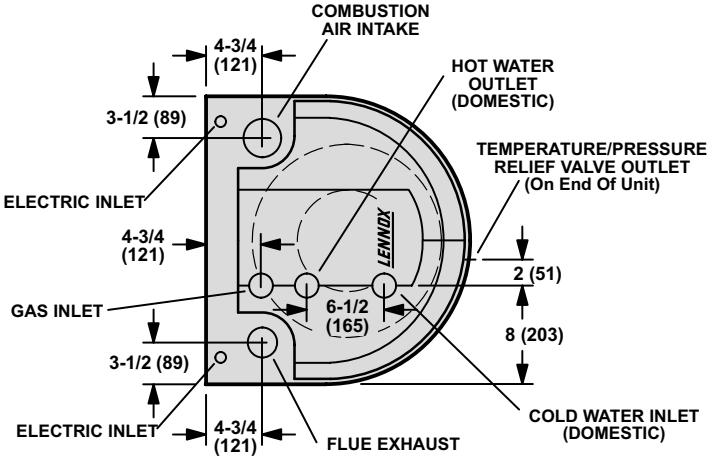

A - Two wire power with ground
B - Three wire control voltage (Heating Only)
Four wire control voltage (Cooling)
C - Four wire control voltage (From Zone 1)
D - Three wire control voltage (For Each Zone) (Up to Four Zones)
E - Four wire control voltage
F - Two wire power with ground (From Zone 1 if close-coupled)
**Or** Separate Disconnect For HM30 If remotely located.
*- Field wiring not furnished -*

All wiring must conform to NEC or CEC and local electrical codes.

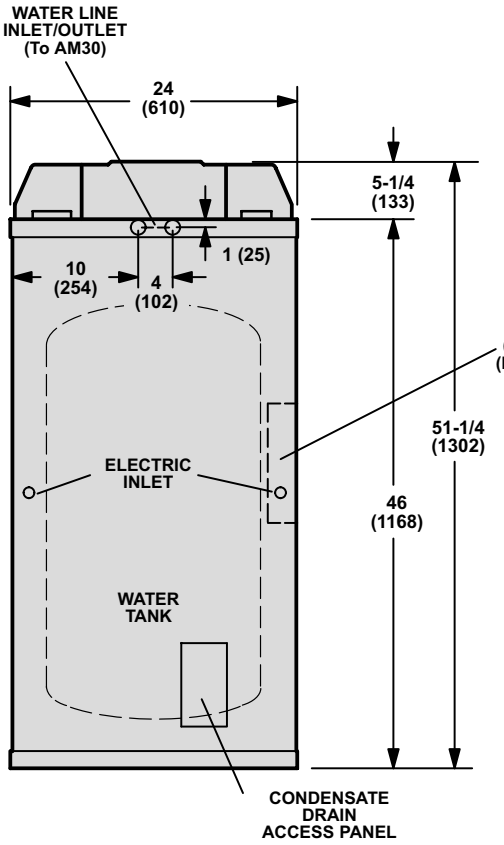
# HM30 DIMENSIONS - INCHES (MM)



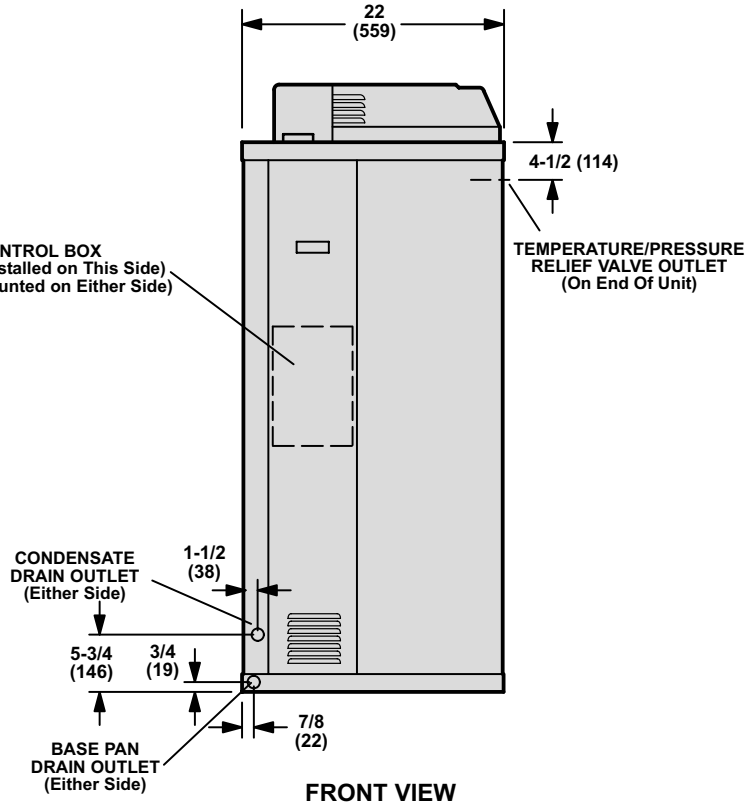

**EXPANSION TANK**




**TOP VIEW**

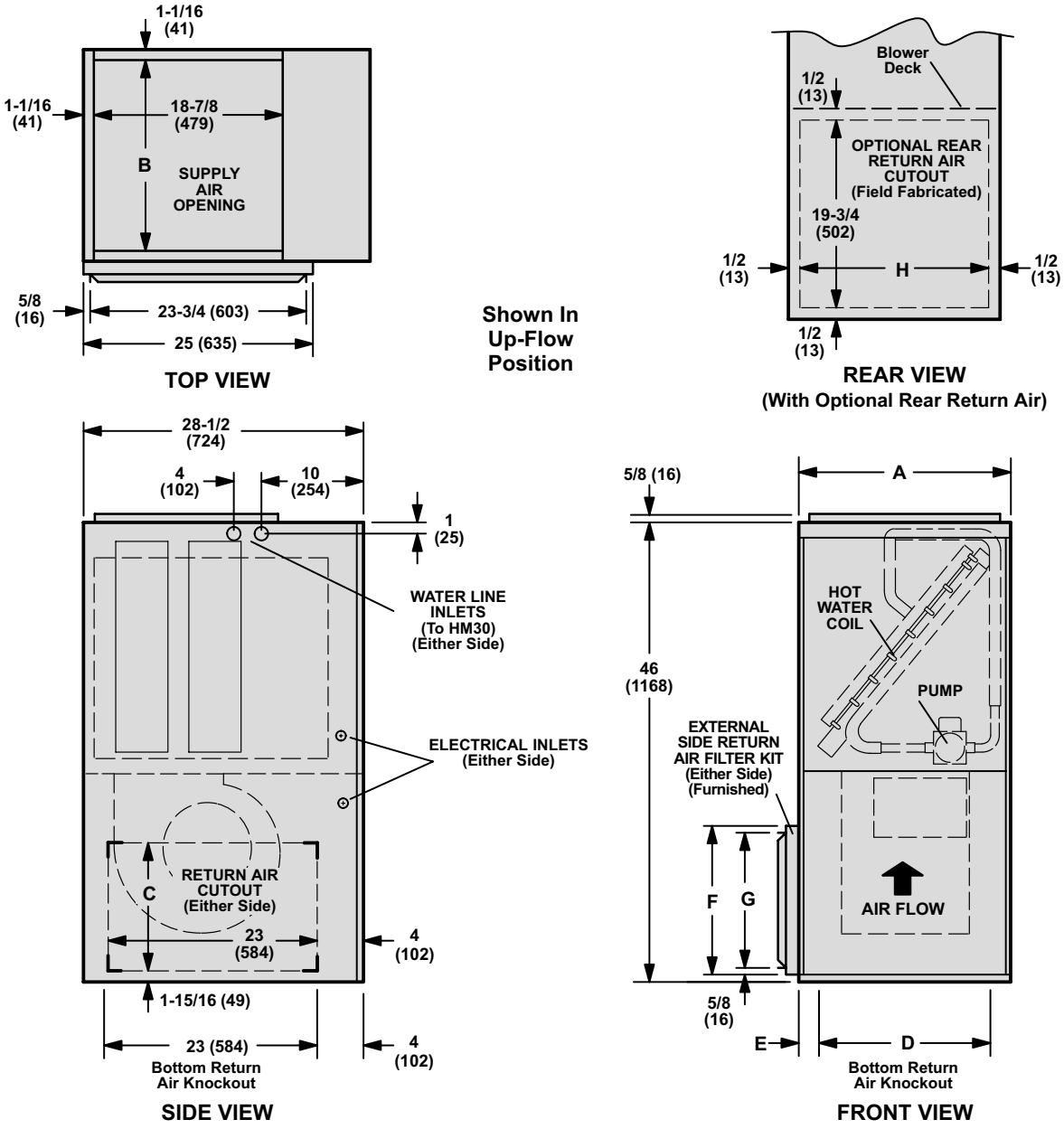



**SIDE VIEW**



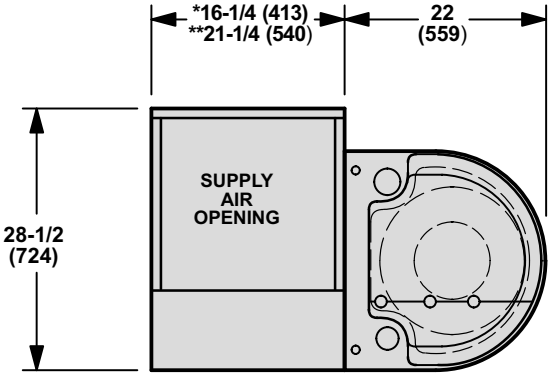

**FRONT VIEW**

# AM30 DIMENSIONS - INCHES (MM)



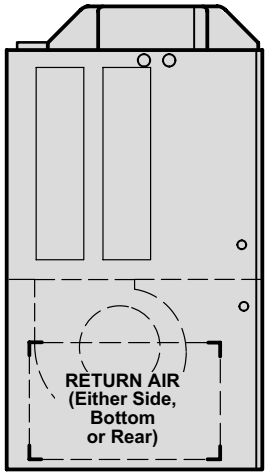

| Model No. | A | | B | | C | | D | | E | | F | | G | | H | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | in. | mm | in. | mm | in. | mm | in. | mm | in. | mm | in. | mm | in. | mm | in. | mm |
| AM30Q2-40<br>AM30Q2/3-70 | 16-1/4 | 413 | 14-1/8 | 359 | 12 | 305 | 12 | 305 | 1-1/8 | 29 | 14 | 356 | 12-3/4 | 324 | 15-1/4 | 387 |
| AM30Q3/4-70<br>AM30Q3/4-90<br>AM30Q4/5-90<br>AM30Q3/4-105<br>AM30Q4/5-120 | 21-1/4 | 540 | 19-1/8 | 486 | 18 | 457 | 18 | 457 | 1-5/8 | 41 | 20 | 508 | 18-3/4 | 476 | 20-1/4 | 513 |

# CLOSE-COUPLED SYSTEM DIMENSIONS (UP-FLOW) - INCHES (MM)

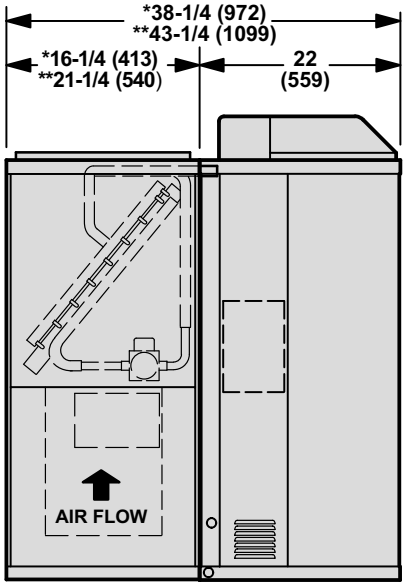



TOP VIEW

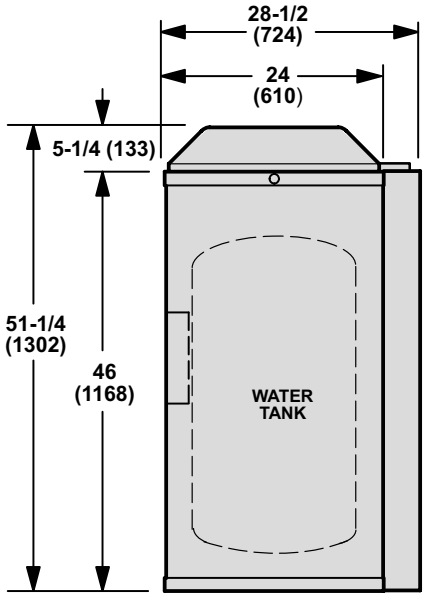
*With AM30Q2-40 Thru AM30Q2/3-70
**With AM30Q3/4-70 Thru AM30Q4/5-120




SIDE VIEW



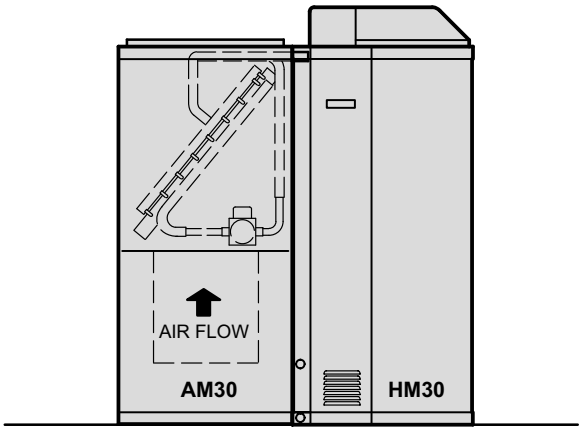

FRONT VIEW



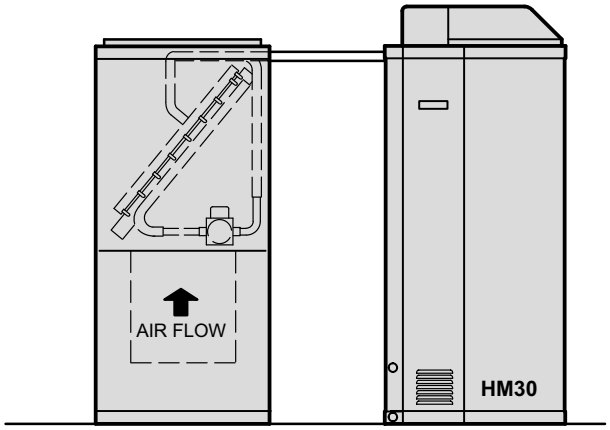

SIDE VIEW

# TYPICAL APPLICATIONS

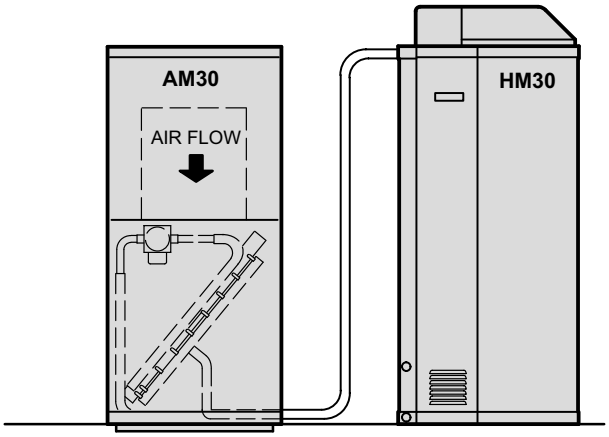
## CompleteHeat Series High Efficiency Gas Heating System Only



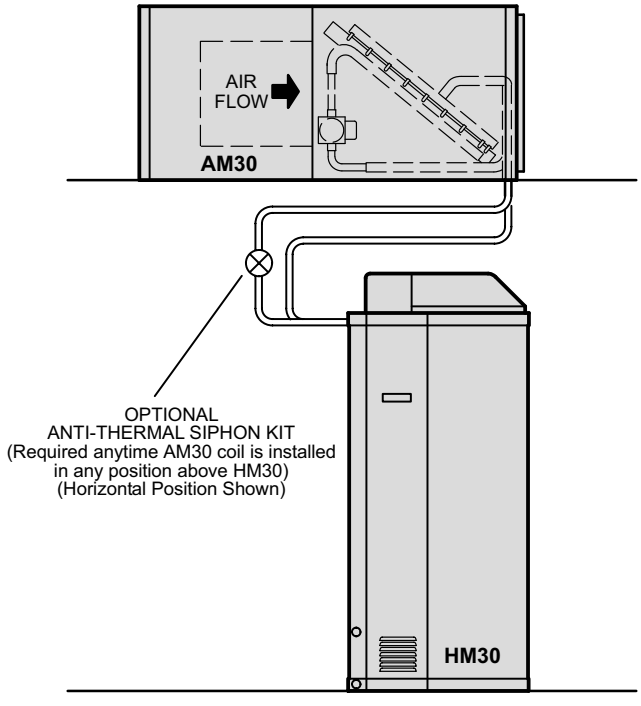

Up Flow Close-Coupled Installation



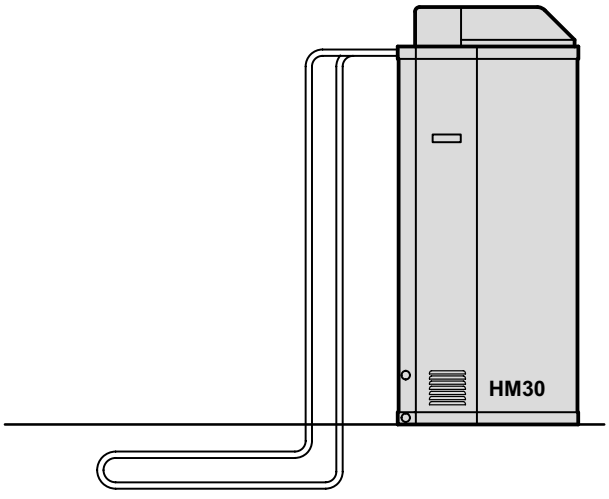

Up Flow Remote Installation




Down Flow Remote Installation

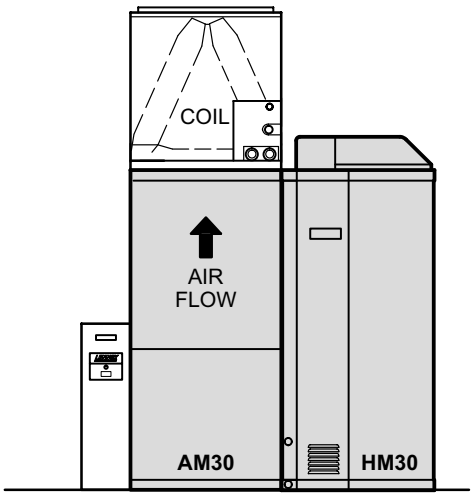



Horizontal Remote Installation

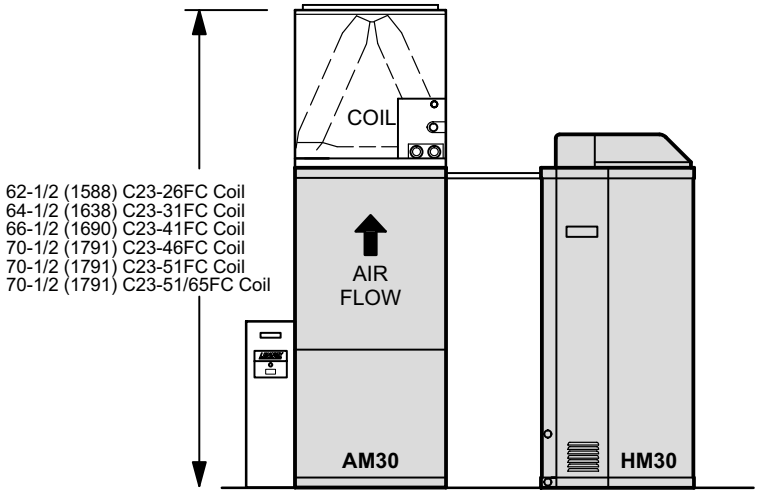



Radiant Floor Heat Installation

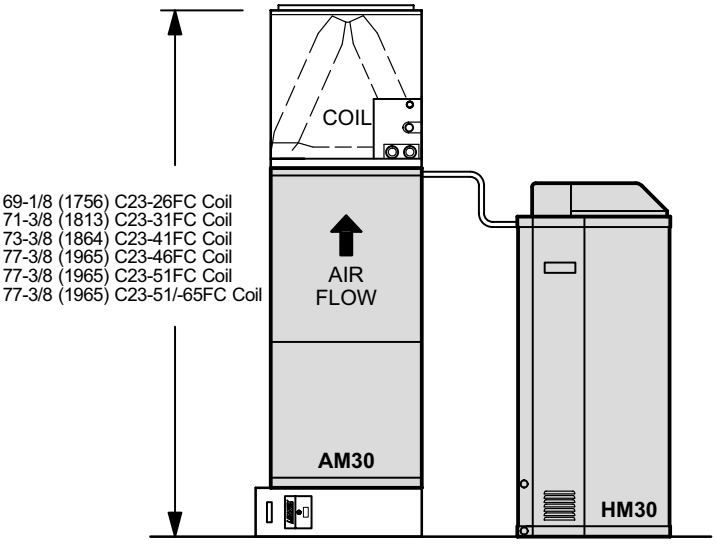
## CompleteHeat Series High Efficiency Gas Heating System With Various Evaporator Coils and Electronic Air Cleaner



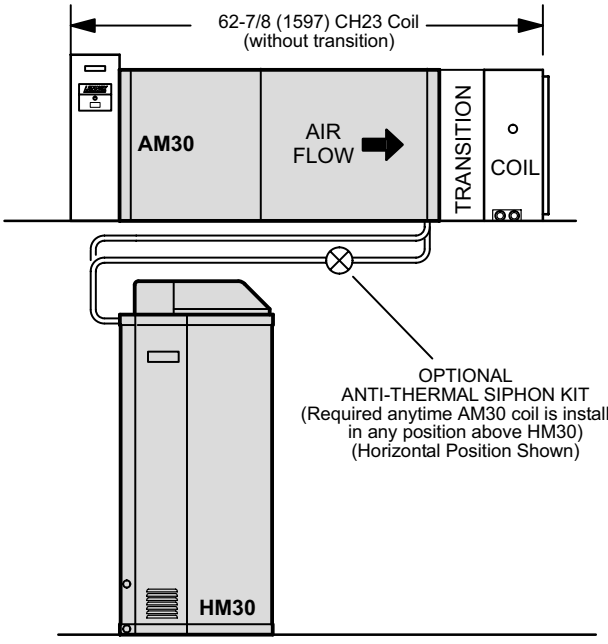

Up Flow Close-Coupled Installation



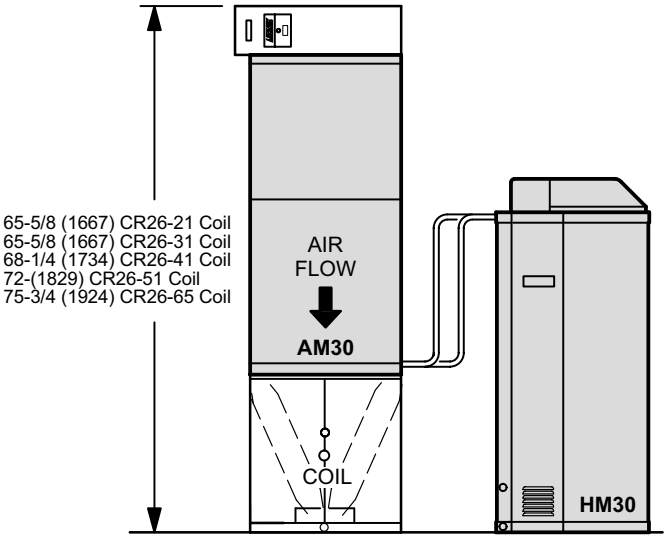

Up Flow Remote Installation




Up Flow Remote Installation



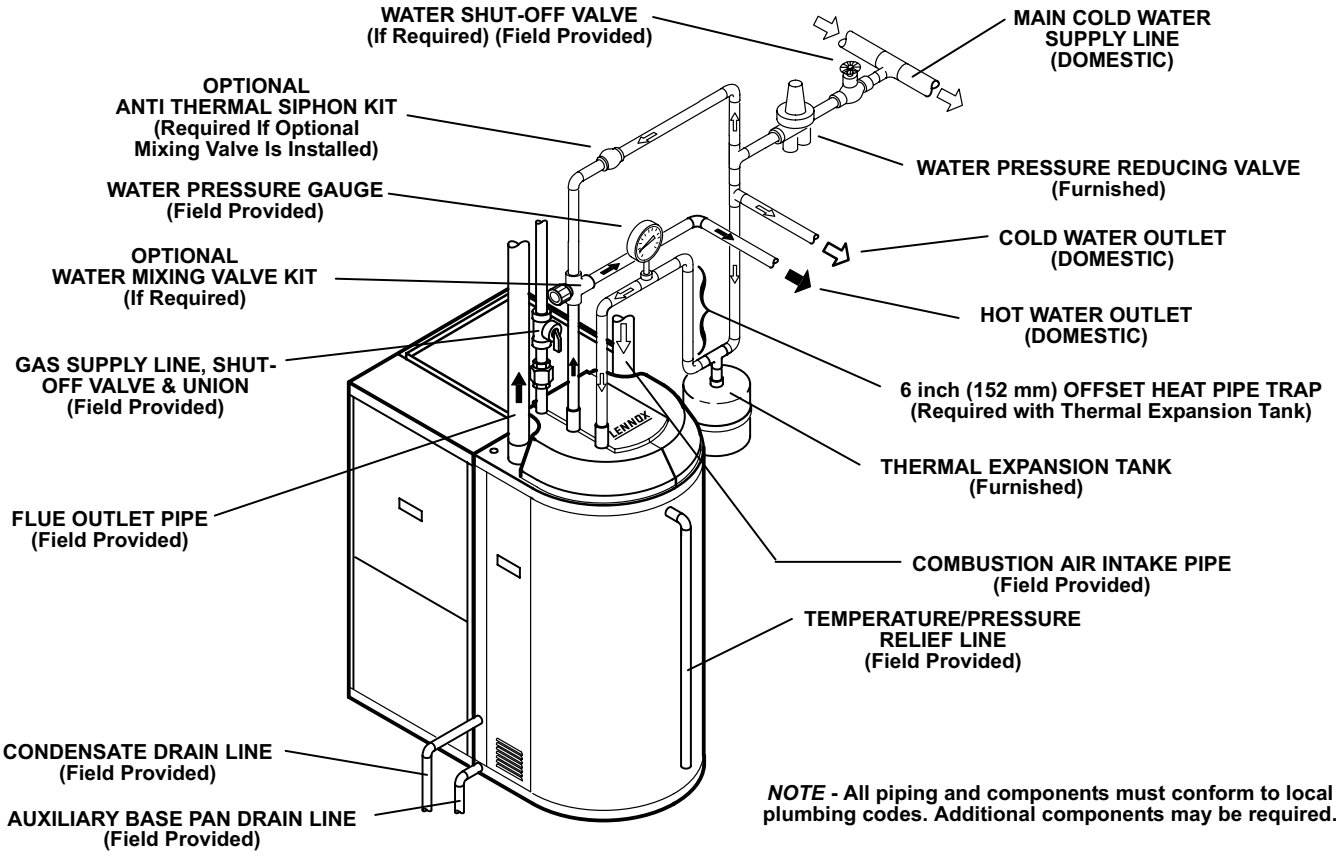

Horizontal Remote Installation




Down Flow Remote Installation

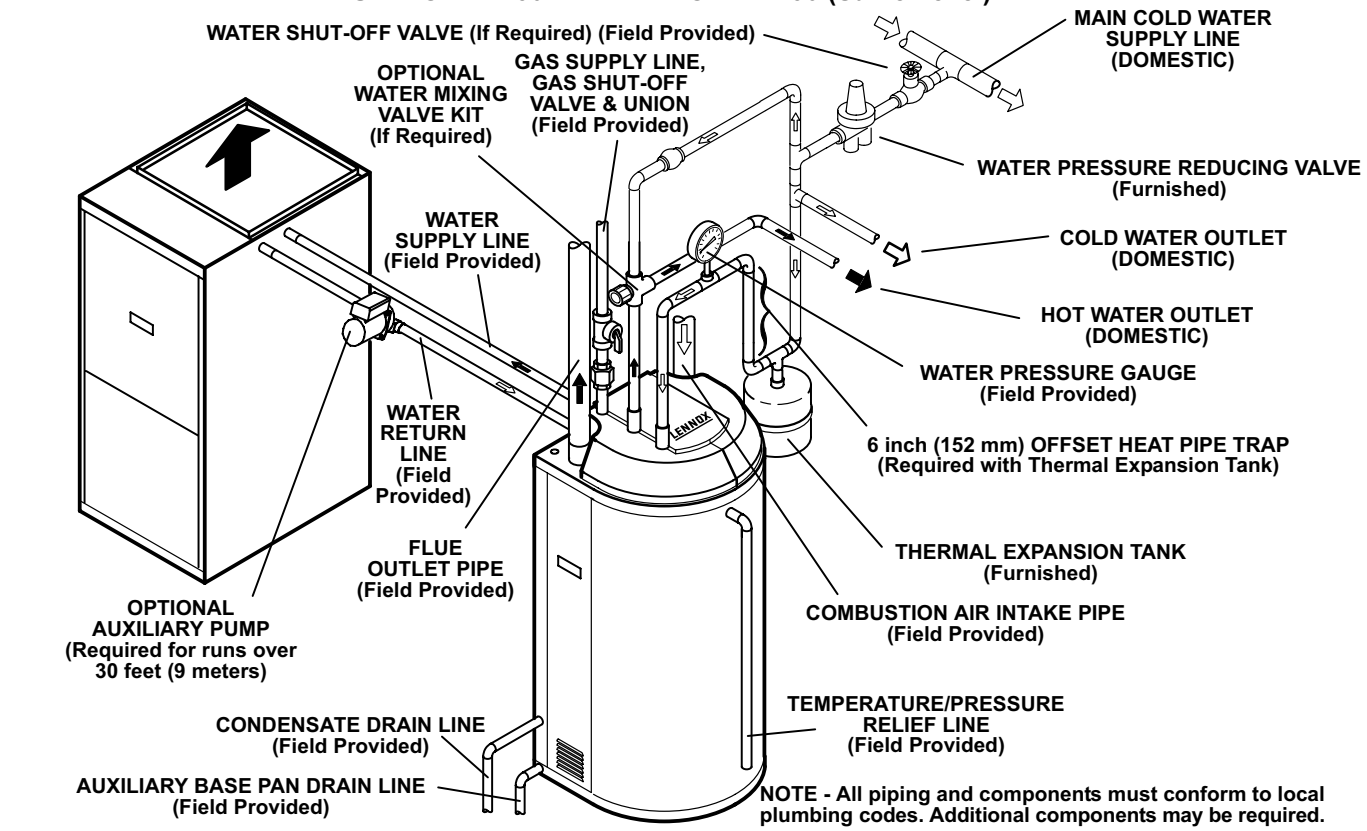
## APPLICATION SCHEMATICS

### UPFLOW AM30 WITH CLOSE-COUPLED HM30




***NOTE*** **- All piping and components must conform to local plumbing codes. Additional components may be required.**

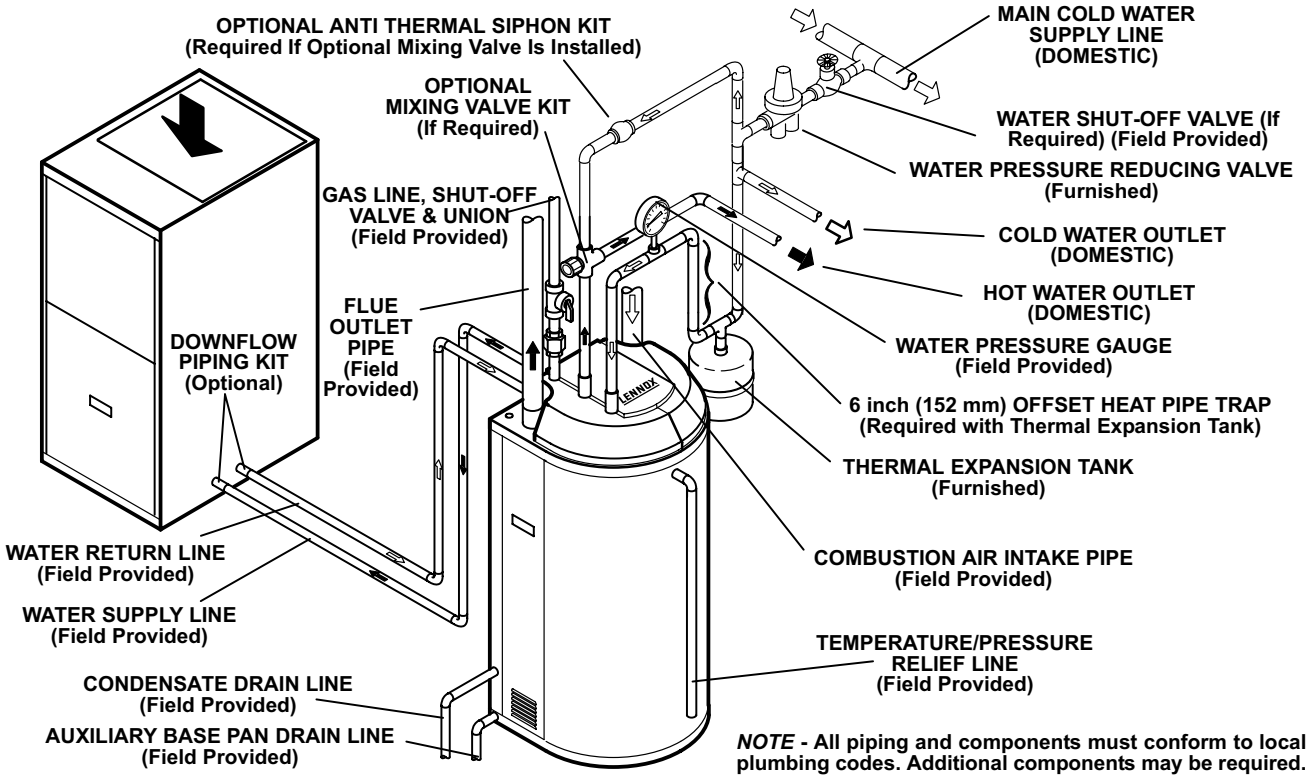
### UPFLOW AM30 WITH REMOTE HM30 (Same Level)




**NOTE - All piping and components must conform to local plumbing codes. Additional components may be required.**

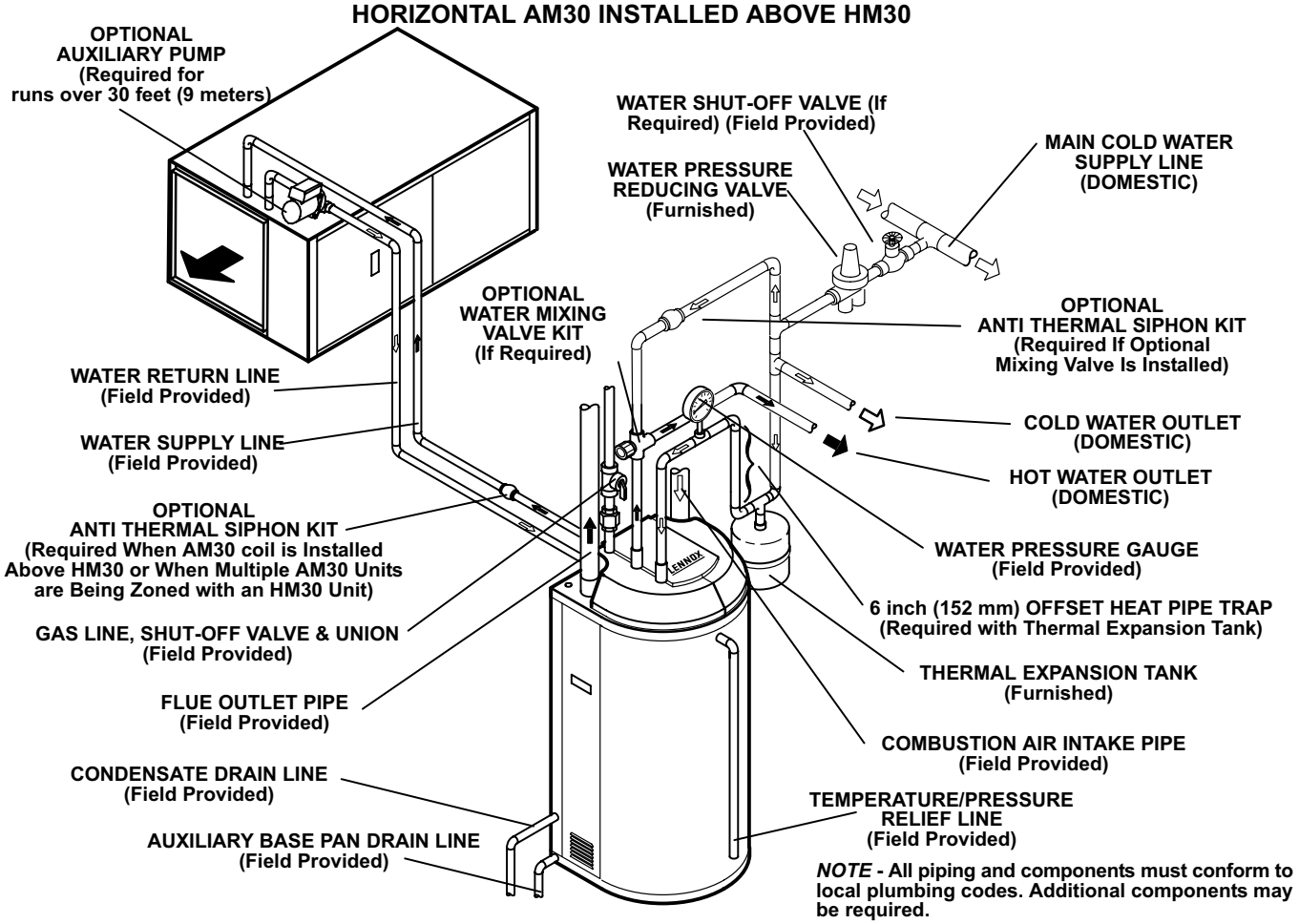
## APPLICATION SCHEMATICS

### DOWNFLOW AM30 WITH REMOTE HM30




***NOTE*** **- All piping and components must conform to local plumbing codes. Additional components may be required.**

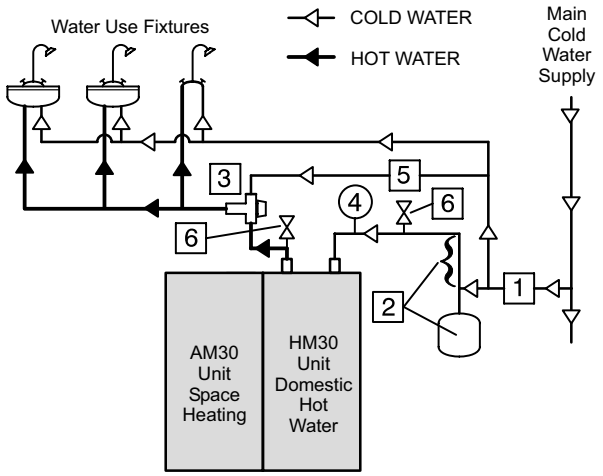
### HORIZONTAL AM30 INSTALLED ABOVE HM30




***NOTE*** **- All piping and components must conform to local plumbing codes. Additional components may be required.**

# PLUMBING SCHEMATICS (ALL CLOSED-LOOP SYSTEMS)

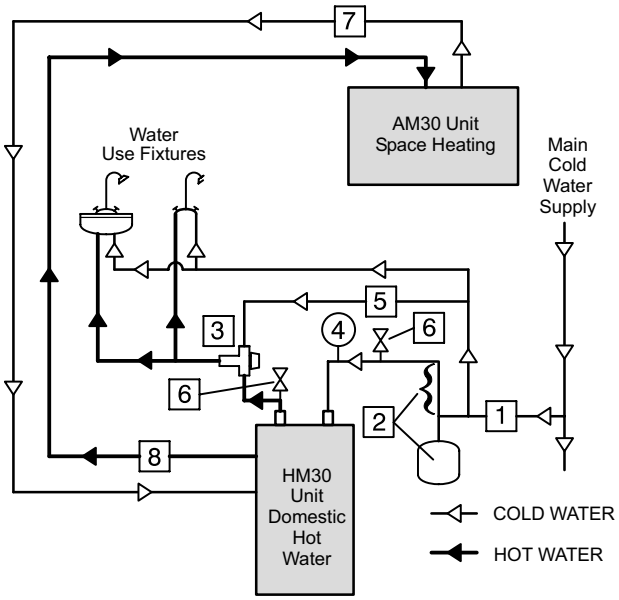
## DOMESTIC HOT WATER and SPACE HEATING (Close-Coupled)




1 - Water Pressure-reducing valve (Required/Furnished)
2 - Thermal expansion tank (Required/Furnished) and 6 inch (152 mm) offset heat pipe trap (Required).
3 - Mixing water valve (if required by local code).
4 - Water pressure gauge - 0 - 150 psi (0 - 1035 kPa).
5 - Anti-thermal siphon kit (check valve). (Required if optional mixing water valve is installed.)
6 - Optional air vent/bleed valve(s). One in cold water line and one in hot water line if line turns down after leaving top of HM30. Installed at high point in line. Valve(s) may be required to bleed air from system.

***NOTE - All plumbing must conform to local codes.***

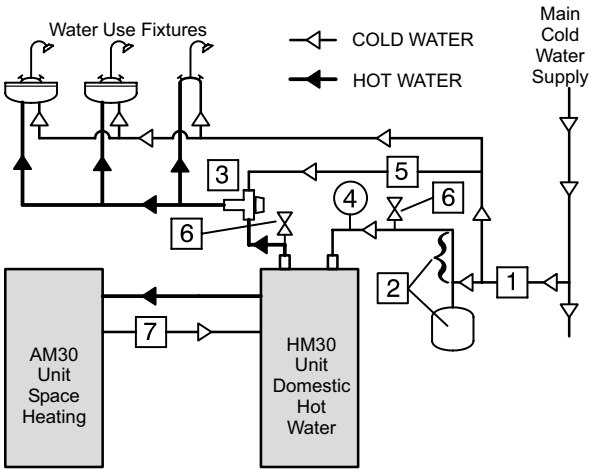
## DOMESTIC HOT WATER and SPACE HEATING (Horizontal)




1 - Water Pressure-reducing valve (Required/Furnished)
2 - Thermal expansion tank (Required/Furnished) and 6 inch (152 mm) offset heat pipe trap. (Required)
3 - Mixing water valve (if required by local code).
4 - Water pressure gauge - 0 - 150 psi (0 - 1035 kPa).
5 - Anti-thermal siphon kit (check valve). (Required if optional mixing water valve is installed.)
6 - Optional air vent/bleed valve(s). One in cold water line and one in hot water line if line turns down after leaving top of HM30. Installed at high point in line. Valve(s) may be required to bleed air from system.
7 - Auxiliary pump. [Required for runs over 30 ft. (9 m).]
8 - Anti-thermal siphon kit (check valve). (Required when AM30 coil is located above HM30.)

***NOTE - All plumbing must conform to local codes.***

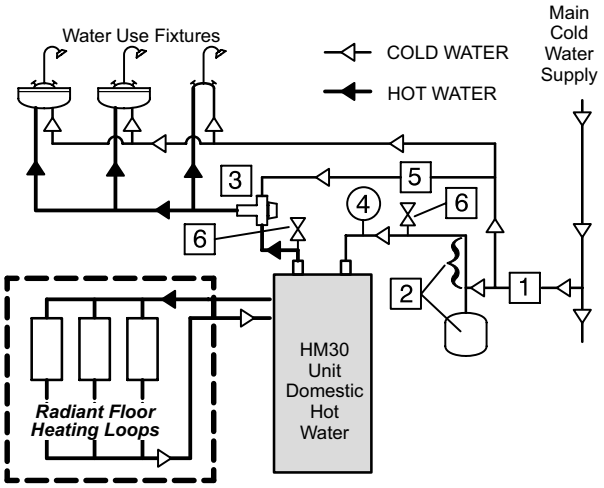
## DOMESTIC HOT WATER and SPACE HEATING (Remote)




1 - Water Pressure-reducing valve (Required/Furnished)
2 - Thermal expansion tank (Required/Furnished) and 6 inch (152 mm) offset heat pipe trap (Required).
3 - Mixing water valve (if required by local code).
4 - Water pressure gauge - 0 - 150 psi (0 - 1035 kPa).
5 - Anti-thermal siphon kit (check valve). (Required if optional mixing water valve is installed.)
6 - Optional air vent/bleed valve(s). One in cold water line and one in hot water line if line turns down after leaving top of HM30. Installed at high point in line. Valve(s) may be required to bleed air from system.
7 - Auxiliary pump. [Required for runs over 30 ft. (9 m).]

***NOTE - All plumbing must conform to local codes.***

## DOMESTIC HOT WATER and RADIANT FLOOR HEATING



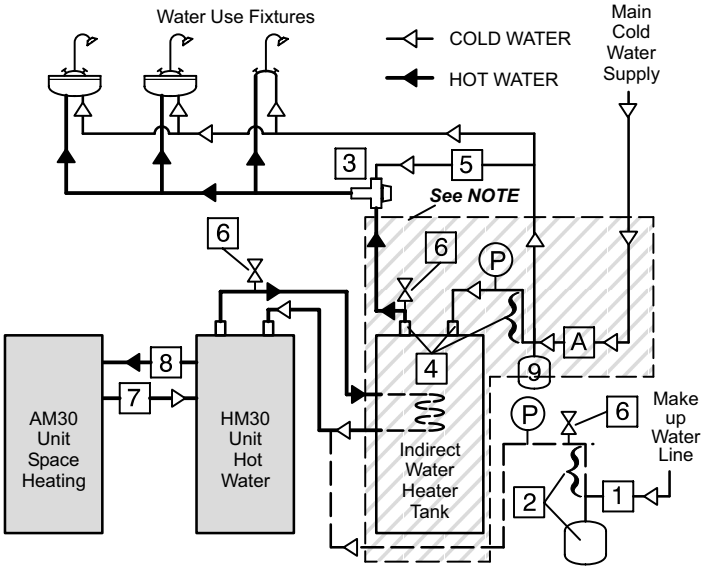

***NOTE - Refer to manufacturer of Radiant Floor Heating Systems for specifications and sizing of components. All piping and components must conform to local plumbing codes. Additional components may be required.***

1 - Water Pressure-reducing valve (Required/Furnished)
2 - Thermal expansion tank (Required/Furnished) and 6 inch (152 mm) offset heat pipe trap (Required).
3 - Mixing water valve (if required by local code).
4 - Water pressure gauge - 0 - 150 psi (0 - 1035 kPa).
5 - Anti-thermal siphon kit (check valve). (Required if optional mixing water valve is installed.)
6 - Optional air vent/bleed valve(s). One in cold water line and one in hot water line if line turns down after leaving top of HM30. Installed at high point in line. Valve(s) may be required to bleed air from system.

***NOTE - All plumbing must conform to local codes.***

# PLUMBING SCHEMATICS (ALL CLOSED-LOOP SYSTEMS)

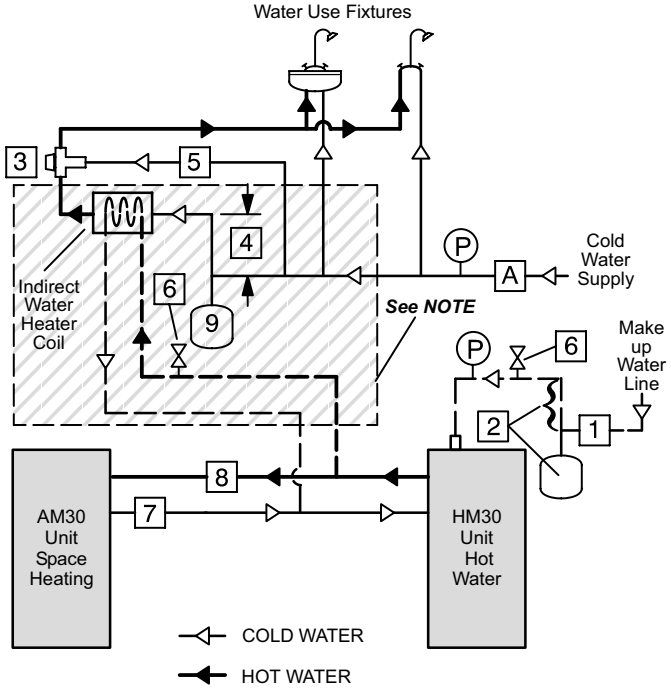
## INDIRECT HEAT TRANSFER DOMESTIC HOT WATER and SPACE HEATING




***NOTE - Refer to manufacturer of Indirect Water Heater for specifications and sizing of components. All piping and components must conform to local plumbing codes. Additional components may be required.***

1 = Water Pressure-reducing valve (Required/Furnished)
2 = Thermal expansion tank (Required/Furnished) and 6 inch (152 mm) offset heat pipe trap. (Required)
3 = Mixing water valve (if required by local code).
4 = Optional in-line heat trap OR 6 inch (152 mm) offset pipe trap may be used to help reduce heat loss due to thermal siphoning. Pipe trap is required when optional Water Mixing Valve is installed.
5 = Anti-thermal siphon kit (check valve). (Required if optional mixing water valve is installed.)
6 = Optional air vent/bleed valve(s). One in cold water line and one in hot water line if line turns down after leaving top of HM30. Installed at high point in line. Valve(s) may be required to bleed air from system.
7 = Auxiliary pump. [Required for runs over 30 ft. (9 m).]
8 = Anti-thermal siphon kit (check valve). (Required when AM30 coil is located above HM30.)
9 = Thermal Expansion Tank (required in closed system)
A= Check Valve, Backflow Preventer, Pressure Reducing Valve or any device which turns the domestic water system into a Closed System. (May be required by local codes. May already be installed in existing water system)
P = Water pressure gauge - 0 - 150 psi (0 - 1035 kPa).

## INDIRECT HEAT TRANSFER (Closed System) DOMESTIC HOT WATER and SPACE HEATING

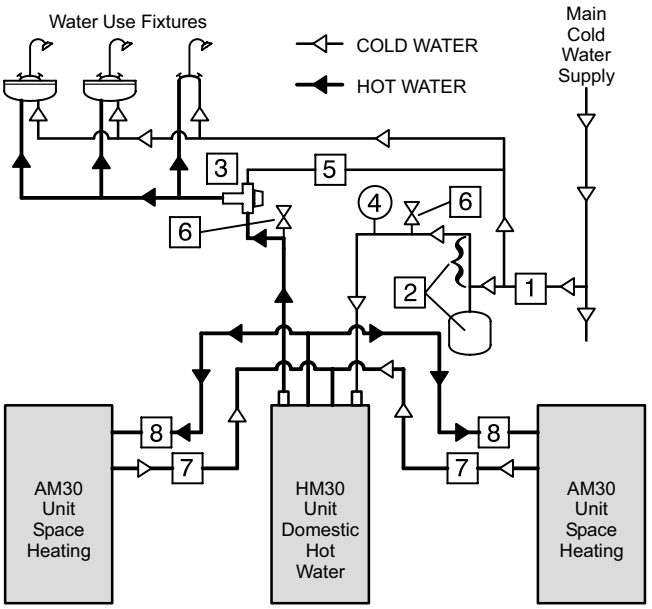



1 = Water Pressure-reducing valve (Required/Furnished)
2 = Thermal expansion tank (Required/Furnished) and 6 inch (152 mm) offset heat pipe trap. (Required)
3 = Mixing water valve (if required by local code).
4 = Optional in-line heat trap OR 6 inch (152 mm) offset pipe trap may be used to help reduce heat loss due to thermal siphoning. Pipe trap is required when optional Water Mixing Valve is installed.
5 = Anti-thermal siphon kit (check valve). (Required if optional mixing water valve is installed.)
6 = Optional air vent/bleed valve(s). One in cold water line and one in hot water line if line turns down after leaving top of HM30. Installed at high point in line. Valve(s) may be required to bleed air from system.
7 = Auxiliary pump. [Required for runs over 30 ft. (9 m).]
8 = Anti-thermal siphon kit (check valve). (Required when AM30 coil is located above HM30.)
9 = Thermal Expansion Tank (required in closed system)
A= Check Valve, Backflow Preventer, Pressure Reducing Valve or any device which turns the domestic water system into a Closed System. (May be required by local codes. May already be installed in existing water system)
P = Water pressure gauge - 0 - 150 psi (0 - 1035 kPa).

***NOTE - Refer to manufacturer of Indirect Water Heater for specifications and sizing of components. All piping and components must conform to local plumbing codes. Additional components may be required.***

# PLUMBING SCHEMATICS (ALL CLOSED-LOOP SYSTEMS)

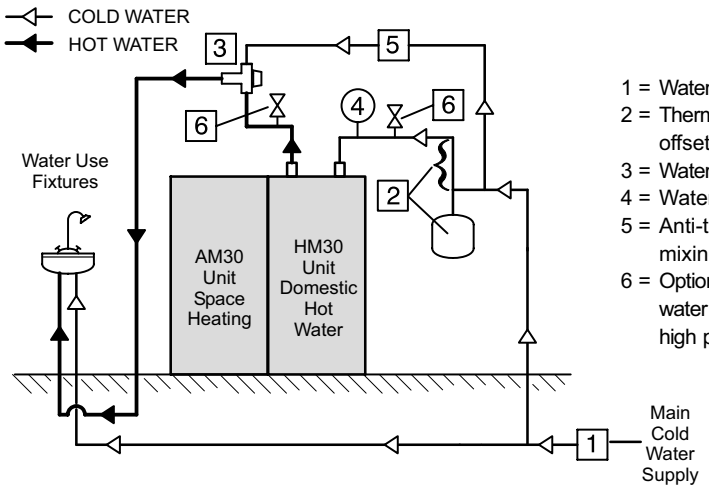
## DOMESTIC HOT WATER and SPACE HEATING
## ONE HM30 HEATING MODULE WITH TWO AM30 AIR MODULES




1 = Water Pressure Reducing Valve (Required/Furnished)
2 = Thermal Expansion Tank (Required/Furnished) and 6 inch (152 mm) offset heat pipe trap (Required)
3 = Water Mixing Valve Kit (if required by local code)
4 = Water pressure gauge - 0 - 150 psi (0 - 1035 kPa).
5 = Anti-thermal siphon kit (check valve). (Required if optional mixing water valve is installed.)
6 = Optional Air Vent/Bleed Valve(s). One in cold water line and one in hot water line if line turns down after leaving top of HM30. Installed at high point in line. Valve(s) may be required to bleed air from system.
7 = Auxiliary Pump (required for runs over 30 ft. (9 m)
8 = Anti-Thermal Siphon Kit (required when AM30 coil is located above HM30).

***NOTE - All piping and components must conform to local plumbing codes. Additional components may be required.***

## DOMESTIC HOT WATER and SPACE HEATING (Close Coupled)
## HOT WATER LINE TURNS DOWN AFTER LEAVING TOP OF HM30

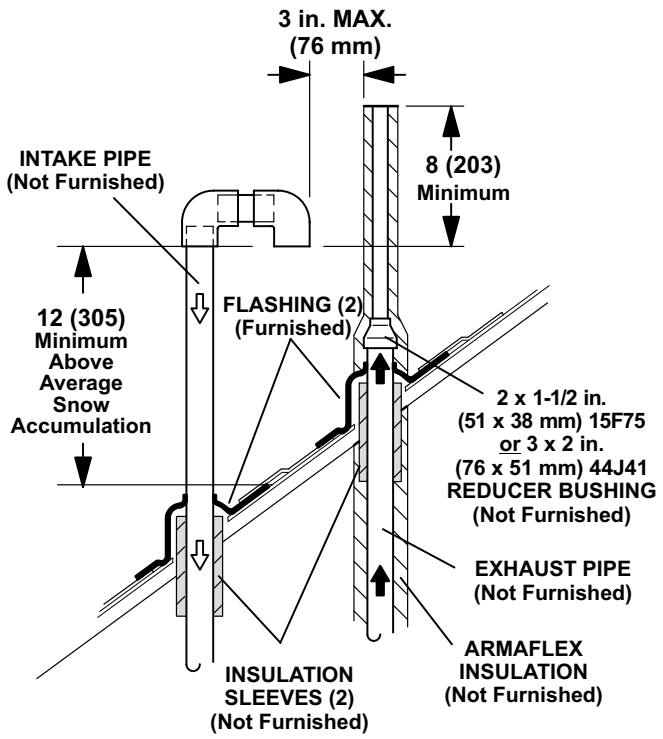



1 = Water Pressure Reducing Valve (Required/Furnished)
2 = Thermal Expansion Tank (Required/Furnished) and 6 inch (152 mm) offset heat pipe trap (Required)
3 = Water Mixing Valve Kit (if required by local code)
4 = Water pressure gauge - 0 - 150 psi (0 - 1035 kPa).
5 = Anti-thermal siphon kit (check valve). (Required if optional mixing water valve is installed.)
6 = Optional Air Vent/Bleed Valve(s). One in cold water line and one in hot water line if line turns down after leaving top of HM30. Installed at high point in line. Valve(s) may be required to bleed air from system.

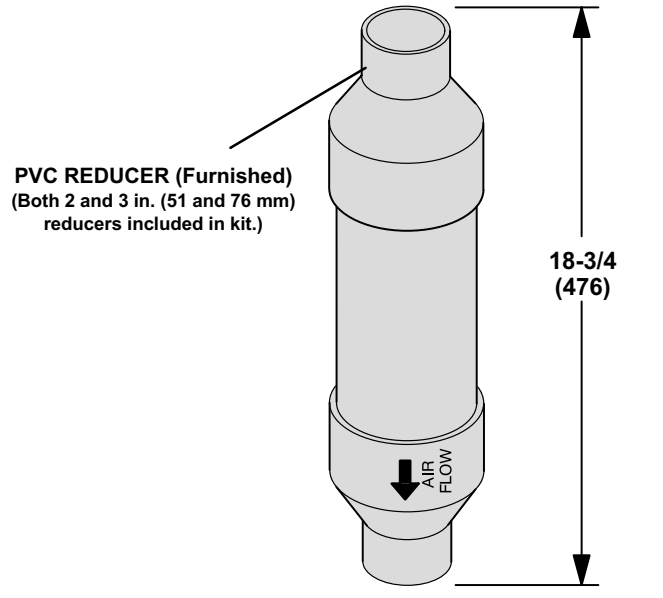
***All plumbing must conform to local codes.***

## OPTIONAL ACCESSORY DIMENSIONS - INCHES (MM)

### ROOF TERMINATION KITS
**15F75 for 2 inch (51 mm) Venting**
**44J41 for 3 inch (76 mm) Venting**

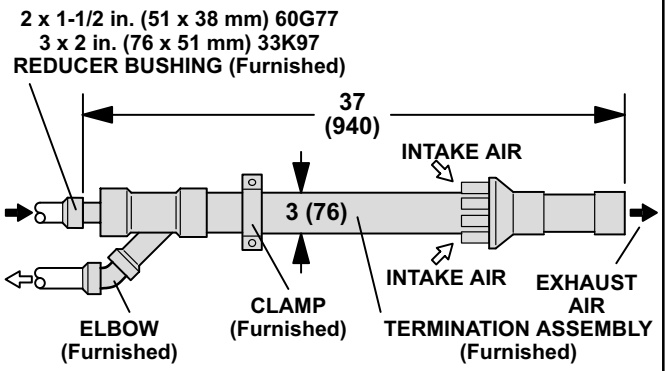



### AIR INTAKE FILTER

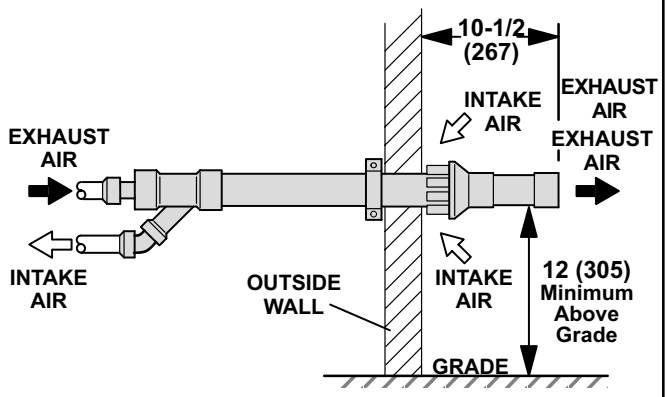



### CONCENTRIC TERMINATION KITS

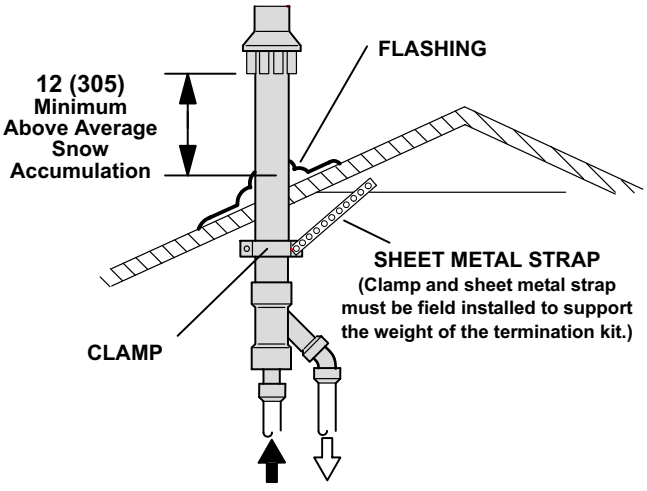
**60G77 for 1-1/2 inch (38 mm) Venting**
**33K97 for 2 inch (51 mm) Venting**



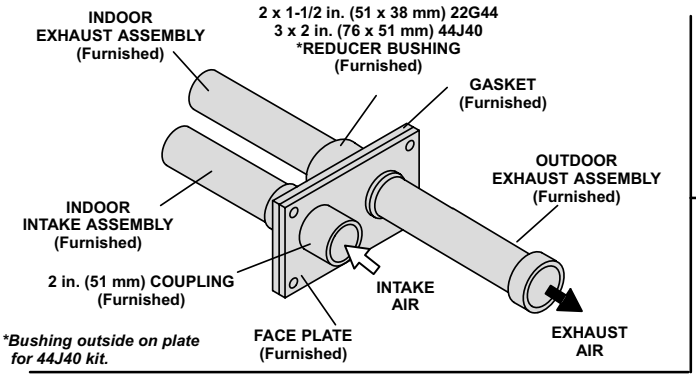

### CONCENTRIC WALL TERMINATION APPLICATIONS




### CONCENTRIC ROOF TERMINATION APPLICATIONS

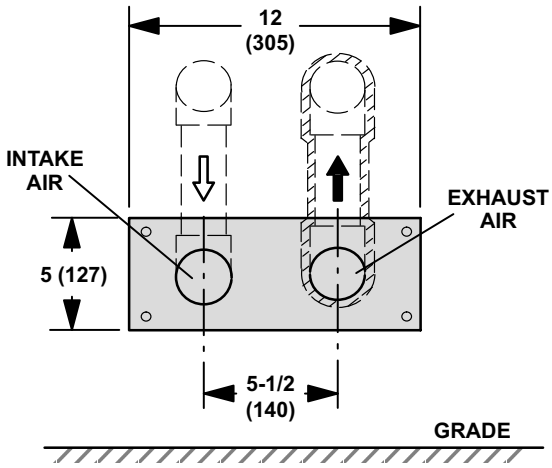


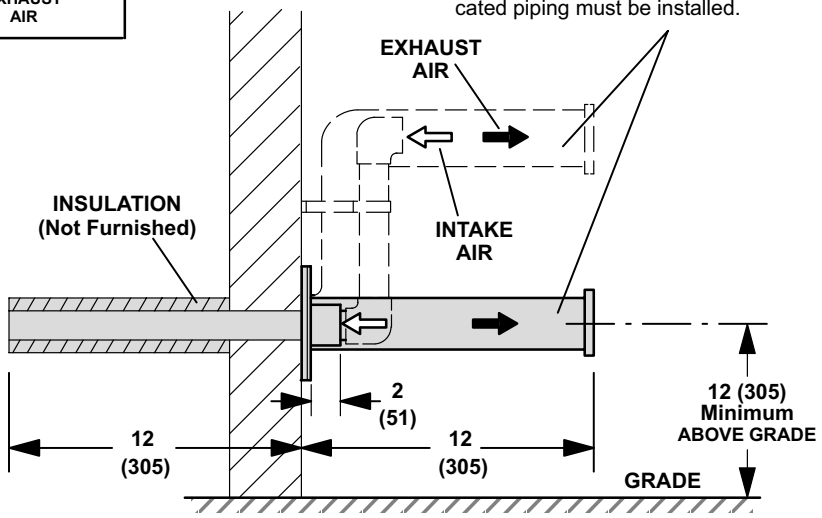
# OPTIONAL ACCESSORY DIMENSIONS - INCHES (MM)

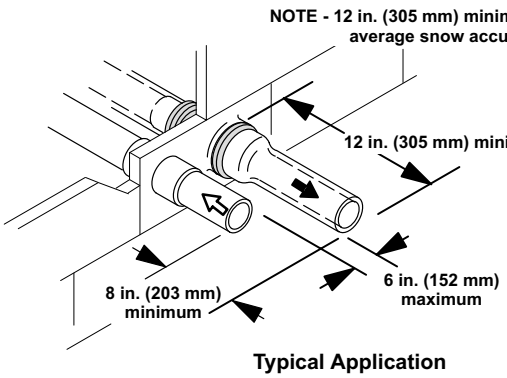



## WALL TERMINATION KITS
## (CLOSE-COUPLE)
## 22G44 for 2 inch (51 mm) Venting
## 44J40 for 3 inch (76 mm) Venting

If Intake and Exhaust Pipe is less than 12 in. (305 mm) above snow accumulation or other obstructions, field fabricated piping must be installed.



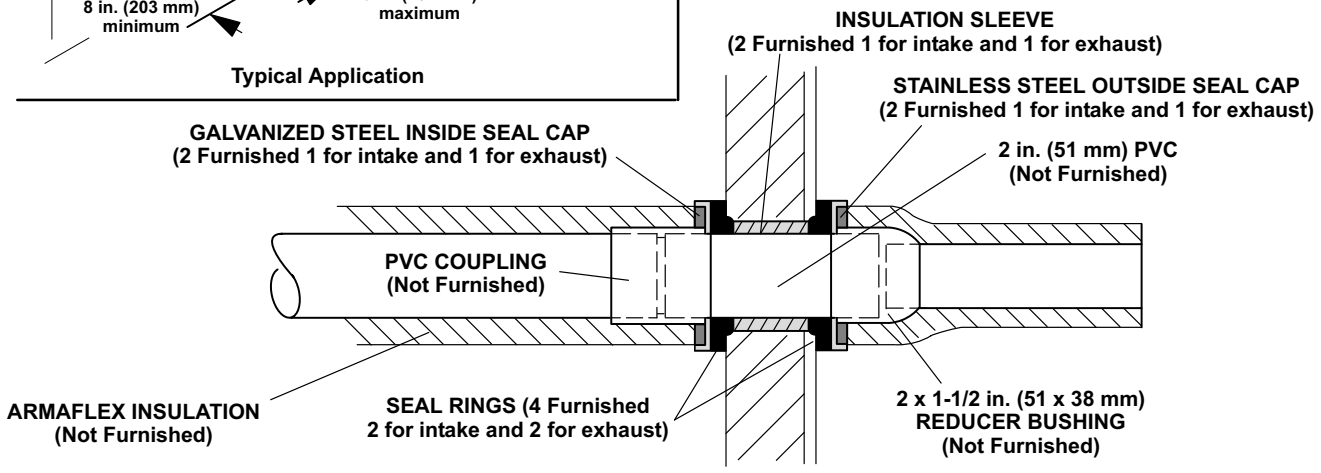




Typical Application

## WALL ASSEMBLY TERMINATION KIT
## (RING KIT)
## 15F74 for 2 inch (51 mm) Venting
## NOTE - Not for use with 3 inch (76 mm) Venting

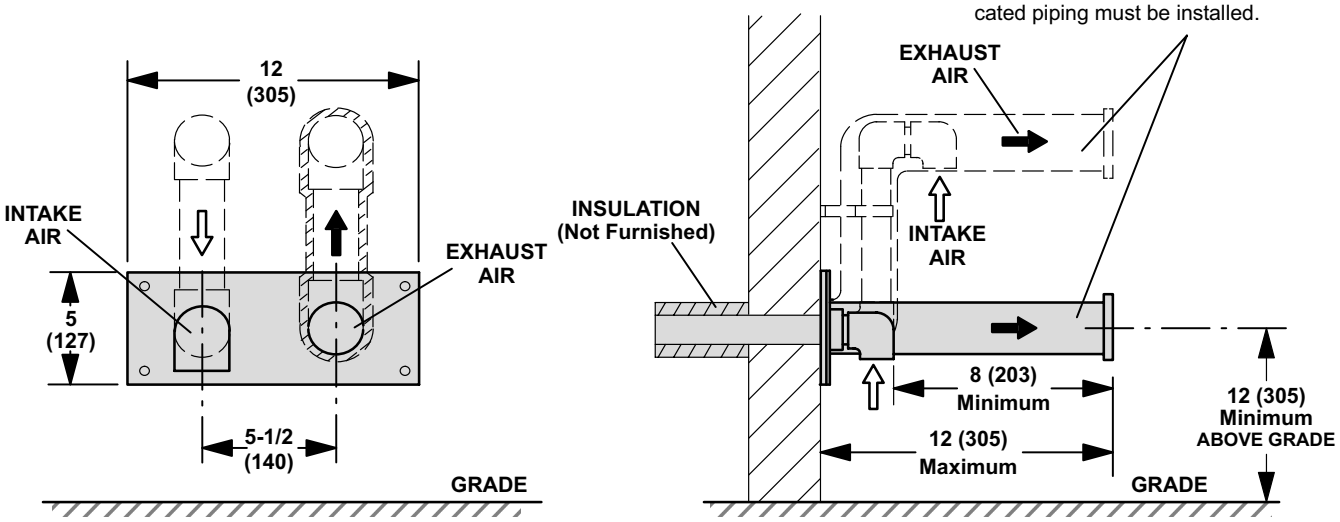
**NOTE - EXHAUST PIPE SHOWN**

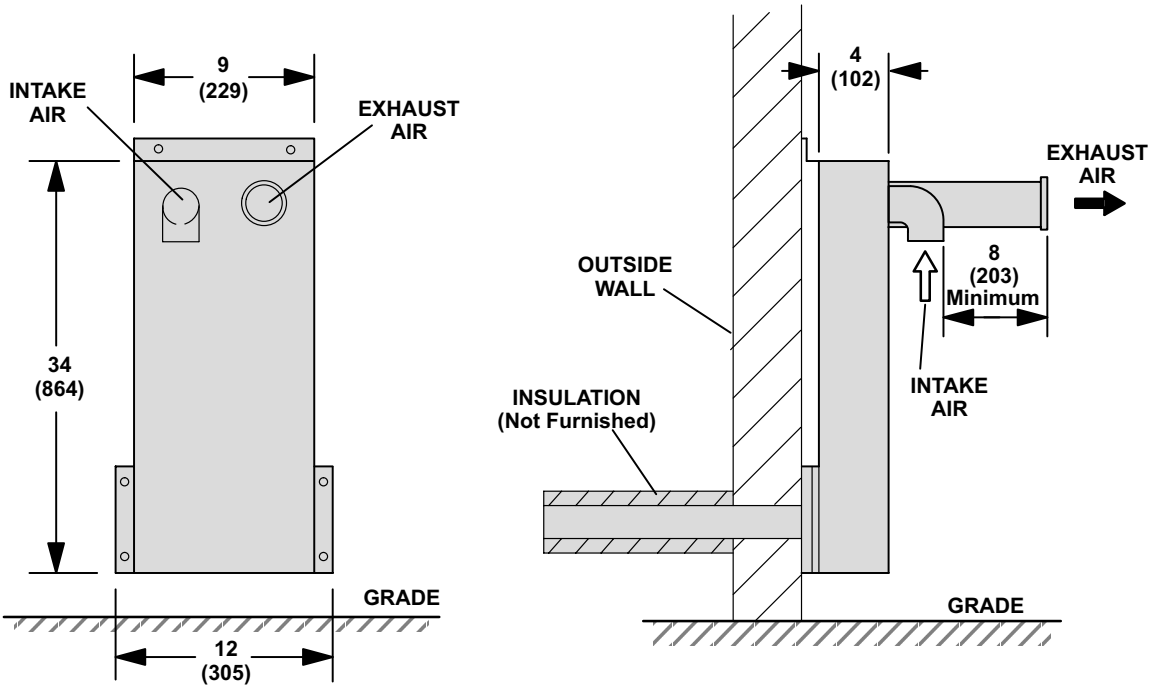


## OPTIONAL ACCESSORY DIMENSIONS - INCHES (MM)

### WTK WALL ASSEMBLY TERMINATION KIT WITH FIELD FABRICATION ABOVE GRADE EXTENDED CLEARANCE
**30G28 for 2 inch (51 mm) Venting**
**81J20 for 3 inch (76 mm) Venting**

If Intake and Exhaust Pipe is less than 12 in. (305 mm) above snow accumulation or other obstructions, field fabricated piping must be installed.




### WTKX WALL ASSEMBLY EXTENSION RISER TERMINATION KIT
**30G79 for 2 inch (51 mm) Venting**
**NOTE - Not for 3 inch (76 mm) Venting**